ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF MEDICAL AND ALLIED SCIENCES

(RSMAS)

DEPARTMENT OF FOOD TECHNOLOGY

COURSE STRUCTURE & SYLLABUS

(BASED ON NATIONAL EDUCATION POLICY 2020)

FOR

B.Sc. IN FOOD TECHNOLOGY

(4 YEARS SINGLE MAJOR)

W.E.F

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TABLE OF CONTENTS

Sl No.	Contents	Page No
1	Preamble	**3**
2	Introduction	**5**
3	Learning Outcomes Based Approach to Curriculum Planning	**6**
4	Award of Degree	**7**
5	Graduate Attributes	**9**
6	Programme Learning Outcomes	**11**
7	Programme Specific Outcomes	**12**
10	Teaching Learning Process	**12**
11	Assessment Methods	**13**
12	Program Structure	**14**
13	Detailed syllabus of Semester-I	**20**
14	Detailed syllabus of Semester-II	**28**
15	Detailed syllabus of Semester-III	**38**
16	Detailed syllabus of Semester-IV	**47**
17	Detailed syllabus of Semester-V	**57**
18	Detailed syllabus of Semester-VI	**65**
19	Detailed syllabus of Semester-VII	**75**
20	Detailed syllabus of Semester-VIII	**85**

1. Preamble

The National Education Policy (NEP) 2020 conceives a new vision for India's higher education system. It recognizes that higher education plays an extremely important role in promoting equity, human as well as societal well-being and in developing India as envisioned in its Constitution. It is desired that higher education will significantly contribute towards sustainable livelihoods and economic development of the nation as India moves towards becoming a knowledge economy and society.

If we focus on the 21st century requirements, the higher education framework of the nation must aim to develop good, thoughtful, well-rounded, and creative individuals and must enable an individual to study one or more specialized areas of interest at a deep level, and also develop character, ethical and Constitutional values, intellectual curiosity, scientific temper, creativity, spirit of service, and twenty-first-century capabilities across a range of disciplines including sciences, social sciences, arts, humanities, languages, as well as professional, technical, and vocational subjects. A quality higher education should be capable enough to enable personal accomplishment and enlightenment, constructive public engagement, and productive contribution to the society. Overall, it should focus on preparing students for more meaningful and satisfying lives and work roles and enable economic independence.

Towards the attainment of holistic and multidisciplinary education, the flexible curricula of the University will include credit-based courses, projects in the areas of community engagement and service, environmental education, and value-based education. As part of holistic education, students will also be provided with opportunities for internships with local industries, businesses, artists, crafts persons, and so on, as well as research internships with faculty and researchers at the University, so that students may actively engage with the practical aspects of their learning and thereby improve their employability.

The undergraduate curriculums are diverse and have varied subjects to be covered to meet the needs of the programs. As per the recommendations from the UGC, introduction of courses related to Indian Knowledge System (IKS) is being incorporated in the curriculum structure which encompasses all of the systematized disciplines of Knowledge which were developed to a high degree of sophistication in India from ancient times and all of the traditions and practises that the various communities of India—including the tribal communities—have evolved, refined and preserved over generations, like for example Vedic Mathematics, Vedangas, Indian Astronomy, Fine Arts, Metallurgy, etc.

At RGU, we are committed that at the societal level, higher education will enable each student to develop themselves to be an enlightened, socially conscious, knowledgeable, and skilled citizen who can find and implement robust solutions to its own problems. For the students at the University, Higher education is expected to form the basis for knowledge creation and innovation thereby contributing to a more vibrant, socially engaged, cooperative community leading towards a happier, cohesive, cultured, productive, innovative, progressive, and prosperous nation."

1.1 INTRODUCTION

The National Education Policy (NEP) 2020 clearly indicates that higher education plays an extremely important role in promoting human as well as societal well-being in India. As envisioned in the 21st-century requirements, quality higher education must aim to develop good, thoughtful, well-rounded, and creative individuals. According to the new education policy, assessments of educational approaches in undergraduate education will integrate the humanities and arts with Science, Technology, Engineering and Mathematics (STEM) that will lead to positive learning outcomes. This will lead to develop creativity and innovation, critical thinking and higher-order thinking capacities, problem-solving abilities, teamwork, communication skills, more in-depth learning, and mastery of curricula across fields, increases in social and moral awareness, etc., besides general engagement and enjoyment of learning. and more in-depth learning.

The NEP highlights that the following fundamental principles that have a direct bearing on the curricula would guide the education system at large, viz.

i. Recognizing, identifying, and fostering the unique capabilities of each student to promote her/his holistic development.

ii. Flexibility, so that learners can select their learning trajectories and programmes, and thereby choose their own paths in life according to their talents and interests.

iii. Multidisciplinary and holistic education across the sciences, social sciences, arts, humanities, and sports for a multidisciplinary world.

iv. Emphasis on conceptual understanding rather than rote learning, critical thinking to encourage logical decision-making and innovation; ethics and human & constitutional values, and life skills such as communication, teamwork, leadership, and resilience.

v. Extensive use of technology in teaching and learning, removing language barriers, increasing access for Divyang students, and educational planning and management.

vi. Respect for diversity and respect for the local context in all curricula, pedagogy,and policy.

vii. Equity and inclusion as the cornerstone of all educational decisions to ensure that all students can thrive in the education system and the institutional environment are responsive to differences to ensure that high-quality education is available for all.

viii. Rootedness and pride in India, and its rich, diverse, ancient, and modern culture, languages, knowledge systems, and traditions.

1.2 Learning Outcomes-Based Approach to Curricular Planning:

The learning outcomes-based curriculum framework is based on the premise that every student and graduate is unique. Each student or graduate has his/her own characteristics in terms of previous learning levels and experiences, life experiences, learning styles and approaches to future career- 2 related actions. The quality, depth and breadth of the learning experiences made available to the students while at the higher education institutions help develop their characteristic attributes.

The key objectives that underpin curriculum planning and development at the undergraduate level include Programme Learning Outcomes, and Course Learning Outcomes. For the B.Sc. (H) Food Technology course it includes:

- To demonstrate comprehensive knowledge and understanding of the food technology curriculum.
- To apply the principles of food science to preserve, process and package to assure the quality and safety of food products.
- To understand that the real-world problems in the food industry requires continuous acquisition of knowledge and its application to improve the safety and quality of a given food or process.
- To analyse, interpret and draw conclusions from quantitative/qualitative data; and critically evaluate ideas, evidence and experiences from an open-minded and reasoned perspective.
- To acquire knowledge and skills, including “learning how to learn“, that are necessary for participating in learning activities throughout life, through self-paced and self-directed learning aimed at personal development, and adapting to changing trades and demands of work place through knowledge/skill development/reskilling.
- To use ICT in a variety of learning situations, demonstrate ability to access, evaluate, and use a variety of relevant information sources.
- To acquire professional competency and entrepreneurial skills for economic empowerment.
- To demonstrate the ability to acquire, analyze, interpret and appropriately present laboratory data

2. Award of Degree

The structure and duration of undergraduate programmes of study offered by the University as per NEP 2020 include:

2.1. Undergraduate programmes of either 3 or 4-year duration with Single Major, with multiple entry and exit options, with appropriate certifications:

2.1.1. UG Certificate: Students who opt to exit after completion of the first year and have secured 40 credits will be awarded a UG certificate if, in addition, they complete one vocational course of 4 credits during the summer vacation of the first year. These students are allowed to re-enter the degree programme within three years and complete the degree programme within the stipulated maximum period of seven years.

2.1.2. UG Diploma: Students who opt to exit after completion of the second year and have secured 80 credits will be awarded the UG diploma if, in addition, they complete one vocational course of 4 credits during the summer vacation of the second year. These students are allowed to re-enter within a period of three years and complete the degree programme within the maximum period of seven years.

2.1.3. 3-year UG Degree: Students who will undergo a 3-year UG programme will be awarded UG Degree in the Major discipline after successful completion of three years, securing 120 credits and satisfying the minimum credit requirement.

2.1.4. 4-year UG Degree (Honours): A four-year UG Honours degree in the major discipline will be awarded to those who complete a four-year degree programme with 160 credits and have satisfied the credit requirements as given in Table 6 in Section 5.

2.1.5. 4-year UG Degree (Honours with Research): Students who secure 75% marks and above in the first six semesters and wish to undertake research at the undergraduate level can choose a research stream in the fourth year. They should do a research project or dissertation under the guidance of a Faculty Member of the University. The research project/dissertation will be in the major discipline. The students who secure 160 credits, including 12 credits from a research project/dissertation, will be awarded UG Degree (Honours with Research).

Table: 1: Award of Degree and Credit Structure with ME-ME

Award	Year	Credits to earn	Additional Credits	Re-entry allowed within (yrs)	Years to Complete
UG Certificate	1	40	4	3	7
UG Diploma	2	80	4	3	7
3-year UG Degree (Major)	3	120	x	x	x
4-year UG Degree (Honours)	4	160	x	x	x
4-year UG Degree (Honors with Research):	4	160	Students who secure cumulative 75% marks and above in the first six semesters		

3. Graduate Attributes:

Table: 7: The Learning Outcomes Descriptors and Graduate Attributes

Sl.no.	Graduate Attribute	The Learning Outcomes Descriptors *(The graduates should be able to demonstrate the capability to:)*
GA1	Disciplinary Knowledge	acquire knowledge and coherent understanding of the chosen disciplinary/interdisciplinary areas of study.
GA 2	Complex problem solving	solve different kinds of problems in familiar and non-familiar contexts and apply the learning to real-life situations.
GA 3	Analytical & Critical thinking	apply analytical thought including the analysis and evaluation of policies, and practices. Able to identify relevant assumptions or implications. Identify logical flaws and holes in the arguments of others. Analyse and synthesize data from a variety of sources and draw valid conclusions and support them with evidence and examples.
GA 4	Creativity	create, perform, or think in different and diverse ways about the same objects or scenarios and deal with problems and situations that do not have simple solutions. Think 'out of the box' and generate solutions to complex problems in unfamiliar contexts by adopting innovative, imaginative, lateral thinking, interpersonal skills, and emotional intelligence.
GA 5	Communication Skills	listen carefully, read texts and research papers analytically, and present complex information in a clear and concise manner to different groups/audiences. Express thoughts and ideas effectively in writing and orally and communicate with others using appropriate media.

GA 6	Research-related skills	develop a keen sense of observation, inquiry, and capability for asking relevant/ appropriate questions. Should acquire the ability to problematize, synthesize and articulate issues and design research proposals, define problems, formulate appropriate and relevant research questions, formulate hypotheses, test hypotheses using quantitative and qualitative data, establish hypotheses, make inferences based on the analysis and interpretation of data, and predict cause-and-effect relationships. Should develop the ability to acquire the understanding of basic research ethics and skills in practicing/doing ethics in the field/ in personal research work.
GA 7	Collaboration	work effectively and respectfully with diverse teams in the interests of a common cause and work efficiently as a member of a team.
GA 8	Leadership readiness/qualities	plan the tasks of a team or an organization and setting direction by formulating an inspiring vision and building a team that can help achieve the vision.
GA 9	Digital and technological skills	use ICT in a variety of learning and work situations. Access, evaluate, and use a variety of relevant information sources and use appropriate software for analysis of data.
GA 10	Environmental awareness and action	mitigate the effects of environmental degradation, climate change, and pollution. Should develop the technique of effective waste management, conservation of biological diversity, management of biological resources and biodiversity, forest and wildlife conservation, and sustainable development and living.

4. Programme Learning Outcomes relating to B. Sc (Honours) degree programme in Food Technology

Students graduating with the degree B.Sc. (Food technology) will be able to achieve the following:

PLO1: Knowledge of Food Technology: Students are able to demonstrate comprehensive knowledge and understanding of one or more disciplines such as chemistry, bio-chemistry, mathematics, statistics, microbiology, engineering, management; regulations with support of different allied subjects of Life Science; Physical Science

PLO2: Develop the ability to solve complex problems Develop: Identify, formulate, review research literature, and analyze complex. FoodTechnology/applications problems and Design solutions for complex problems and design system components or processes that meet the specified needs with appropriate consideration for the food sustainability

PLO3: **Develop Critical thinking and analytical reasoning ability:** Recognize the need for, and have the preparation and ability to engage in independent/as an entrepreneur and life-long learning in the broadest context of technological change logical reasoning and capability of recognizing and distinguishing the various aspects of real-life problems.

PLO4: Develop the ability to create: Recognize new skills, ideas and technologies and its implementation in new product developments**.**

PLO5: Communication Skills: Communicate effectively and write effective reports and design documentation, make effective presentation through seminars, project dissertations

PLO6: Develop Research related skills: Acquire the practical knowledge and demonstrate the ability to design, conduct/trouble shoot experiments and analyze data in the field of food technology

PLO7: Develop the skills for collaborative work and team building: Work effectively with food industries, laboratories and production processing team to build the technical and practical learning aspects.

PLO8: **Develop Leadership qualities:** Work effectively with the team work and building capabilities and leadership qualities for achieving the vision**.**

PLO9: Develop Digital and technological skills: The completion of this programme will enable the learner to useappropriate software's to apply for bulk scale/industrial production of technology-based food products

PLO10: Develop Environmental awareness and imbibe skills for addressing the problems: Examining the role of health consciousness, environmental awareness and intention on purchase of organic food

5. B.Sc. Food Technology Programme Specific Outcomes

The programme specific outcomes of the course are-

PSO 1: Knowledge of various areas related to Food science and technology,
PSO 2: Understanding of the food composition and its physico-chemical, nutritional, microbiologicaland sensory aspects,
PSO 3: Knowledge of processing and preservation techniques of pulses, oilseeds, spices, fruits andvegetables, meat, fish, poultry, milk & milk products,
PSO 4: Relevance and significance of food safety, food quality, food plant sanitation, food laws andregulations, food engineering and packaging in food industry.

6. Teaching Learning Process

Teaching and learning in this programme involve classroom lectures as well as tutorial and remedialclasses.

Tutorial classes: Tutorials allow closer interaction between students and teacher as each student gets individual attention. The tutorials are conducted for students who are unable to achieve average grades in their weekly assessments. Tutorials are divided into three categories, viz. discussion-based tutorials (focusing on deeper exploration of course content through discussions and debates), problem-solving tutorials (focusing on problem solving processes and quantitative reasoning), and Q&A tutorials (studentsask questions about course content and assignments and consolidate their learning in the guiding presenceof the tutor).
Remedial classes: The remedial classes are conducted for students who achieve average and above average grades in their weekly assessments. The focus is laid to equip the students to perform better in theexams/assessments. The students are divided into small groups to provide dedicated learning support. Tutors are assigned to provide extra time and resources to help them understand concepts with advanced nuances. Small groups allow tutors to address their specific needs and monitor them. Following methods are adopted for tutorial and remedial classes:

- Written assignments and projects submitted by students
- Project-based learning
- Group discussions
- Home assignments
- Class tests, quizzes, debates organised in the department
- PPT (Presentation), Seminars and conferences
- Extra-curricular activities like cultural activities, community outreach programmes etc.
- Industrial tour / field visit

7. Assessment Methods

	Component of Evaluation	Marks	Frequency	Code	Weightage (%)
A	**Continuous Evaluation**				
I	Analysis/Class test	Combination of any three from (i) to (v)with 5 marks each	1-3	C	25 %
Ii	Home Assignment		1-3	H	
Iii	Project		1	P	
Iv	Seminar		1-2	S	
V	Viva-Voce/Presentation		1-2	V	
Vi	MSE	MSE shallbe of 10 Marks	1-3	Q/CT	
Vii	Attendance	Attendance shall be of 5marks	100%	A	5%
B	Semester End Examination		1	SEE	70%
	Project				**100%**

STRUCTURE OF THE SYLLABUS FOR 4 YEAR UG PROGRAMME

SCHOOL NAME - ROYAL SCHOOL OF MEDICAL AND ALLIED HEALTH SCIENCES (RSMAS)

DEPARTMENT NAME -DEPARTMENT OF FOOD SCIENCE AND NUTRITION

PROGRAMME NAME -B.Sc. IN FOOD TECHNOLOGY

1ST Semester Syllabus

COMPONENT	COURSE CODE	COURSE TITLE	CREDIT	L-T-P
Major (Core)	FTC152M101	Principle of Food Processing and Preservation	3	**3-0-0**
	FTC152M111	Practical on Food Processing and Preservation	3	**0-0-6**
	FTC152M102	Basic Food Science	3	**3-0-0**
IKS	IKS 1	IKS I (Introduction to Indian Knowledge System-I)	3	**2-1-0**
Ability Enhancement Course (AEC)	AEC982A101	Communicative English	2	**1-0-0**
Skill Enhancement Course (SEC)	FTC152S111	Fruits and Vegetables Processing	3	**0-0-6**
Value Added Course (VAC)	VAC 1	Basket Course	3	**3-0-0**
MOOCs	Swayam/MOOCs	*Swayam/MOOCs course will be identified by the dept. from the list of courses available on the MOOCs/Swayam portal*	3	**3-0-0**
		TOTAL CREDIT	**23**	

		2nd Semester		
COMPONENT	**COURSE CODE**	**COURSE TITLE**	**CREDIT**	**L-T-P**
Major (Core)	FTC152M201	Food Microbiology	3	**3-0-0**
	FTC152M211	Practical on Food Microbiology	2	**0-0-4**
	FTC152M202	Fermentation Technology	3	**3-0-0**
	FTC152M212	Practical on Fermentation Technology	2	**0-0-4**
IKS	IKS II	IKS II (Introduction to Indian Knowledge System - II)	3	**3-0-0**
AEC	AEC982A201	Communicative English	1	**1-0-0**
SEC	FTC152S211	Technique in Bakery	3	**0-0-6**
VAC	VAC 2	Basket Course	3	**3-0-0**
Swayam/MOOCs	Swayam/MOOCs 2	*Swayam/MOOCs course will be identified by the dept. from the list of courses available on the MOOCs/Swayam portal*	3	**—**
		TOTAL CREDIT	**24**	

		3rd SEMESTER		
COMPONENT	**COURSE CODE**	**COURSE TITLE**	**CREDIT**	**L-T-P**
Major (Core)	FTC152M301	Food Chemistry	4	**4-0-0**
	FTC152M302	Cereal, Pulses and Legumes and oilseeds Technology	4	**4-0-0**
	FTC152M311	Practical on Food Chemistry	2	**0-0-4**
	FTC152M312	Practical on Cereal, Pulses and Legumes and oilseeds Technology	2	**0-0-4**
IDC	IDC 3	Basic Food Processing	3	**3-0-0**
AEC	AEC982A301	Communicative English	1	**1-0-0**
SEC	FTC152S311	Waste and by product Utilization	3	**0-0-6**
***Swayam/ MOOCs**	Swayam/ MOOCs 3	*Swayam/MOOCs course will be identified by the dept. from the list of courses available on the MOOCs/Swayam portal	3	
		TOTAL CREDIT	**22**	

		4th Semester		
COMPONENT	**COURSE CODE**	**COURSE TITLE**	**CREDIT**	**L-T-P**
Major (Core)	FTC152M401	Concepts of Food Engineering	4	**4-0-0**
	FTC152M402	Food Product Development	3	**3-0-0**
	FTC152M411	Practical on Concepts of Food Engineering	3	**0-0-6**
	FTC152M403	Food Analysis and Instrumentation	4	**4-0-0**
	FTC152M412	Practical on Food Product Development	4	**0-0-8**
AEC	FTC152M412	Communication English	1	**1-0-0**
MOOCs	BHS982A402	*Swayam/MOOCs course will be identified by the dept. from the list of courses available on the MOOCs/Swayam portal*	3	—
		TOTAL CREDIT	**22**	

		5th Semester		
COMPONENT	**COURSE CODE**	**COURSE TITLE**	**CREDIT**	**L-T-P**
Major (Core)	FTC152M501	Dairy Technology	4	**4-0-0**
	FTC152M502	Animal Product Technology	4	**4-0-0**
	FTC152M503	Food Quality and Sensory Evaluation	4	**4-0-0**
	FTC152M511	Practical on Dairy and Animal Product Technology	4	**0-0-8**
	FTC152M521	**Internship**	4	
		TOTAL CREDIT	**20**	

		6th Semester		
COMPONENT	**COURSE CODE**	**COURSE TITLE**	**CREDIT**	**L-T-P**
Major (Core)	FTC152M601	Food Toxicology	4	**4-0-0**
	FTC152M602	Food Safety, Standard and Quality Control	4	**4-0-0**
	FTC152M603	Unit Operation in Food Processing	4	**4-0-0**
	FTC152M604	Nutraceuticals and Functional Foods	4	**4-0-0**
	FTC152M611	Practical on Food Safety and Standard	4	**0-0-8**
		TOTAL CREDIT	**20**	

	7th Semester					
Sl. No.	**Subject Code**	**Names of Subject**	**L**	**Level**	**P**	**C**
		Major				
1	FTC152M701	Research Methodology	4	400	0	4
2	FTC152M702	Fermentation Technology	4	400	0	4
3	FTC152M711	Practical on Research Methodology and Fermentation Technology	0	400	8	4
4	FTC152M712	Research Project	0		0	4
		Minor				
5	FTC152N701	Food Business Management	4	300	0	4
		TOTAL CREDIT	**16**		**8**	**20**

	8th Semester					
Sl. No.	**Subject Code**	**Names of Subject**	**L**	**Level**	**P**	**C**
		Major				
1	FTC152M801	Nutraceuticals and Functional Foods	4	400	0	4
2	FTC152M802	Concepts of Food Engineering	4	400	0	4
		Minor				
2	FTC152N801	Food Quality and Sensory Evaluation	4	300	0	4
		Dissertation				
3	FTC152M812	Research Project	0		0	8
		OR				
4	FTC152M803	Food Plant Sanitation	4	400	0	4
5	FTC152M804	Food Adulteration and Additives	4	400	0	4
		TOTAL CREDIT	**8**		**0**	**20**

Semester I

Course: Major **Scheme of Evaluation(T)** **L-T-P-C: 3-0-0-3**
Subject Title: Principles of Food Processing and Preservation **Level 100**
Subject Code: FTC152M101 **Total credits:3**

Course Objectives

To train the students with various types of processing techniques used in food industry and to understand how processing can lead to increase in food shelf life and palatability.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Relate** the principle of food processing techniques, its function and application	**BT 1**
CO 2	**Explain** different techniques of freezing in food processing	**BT 2**
CO 3	**Develop** an insight into the different techniques of food	**BT 3**
CO 4	**Analyse** the novel techniques of food preservation	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I.	**Food processing and preservation principles:** Method of preservation: pasteurization, sterilization, blanching, canning, packaging	**15**
II.	**Food Concentration:** Definition of drying and dehydration, Factors affecting drying, types of dryers. Definition of evaporation, types of evaporators. Freeze concentration and food freezing and thawing and the general principles of crystallization and nucleation of ice.	**15**

III.	**Basic unit operations in food processing:** Cleaning, sorting, grading blanching, cutting, dicing, mincing, slicing, chopping, grating sieving, soaking, coating, germination and fermentation, filtration techniques	**15**
IV.	**Introduction to Novel Food Preservation Techniques:** Basics of Ohmic Heating, irradiation, pulsed electric field, pulsed light, high pressure processing, cold plasma	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Textbooks:

- Potter NH,5th edition, Food Science, CBS Publication, New Delhi.
- RamSaswamy H and Marcotte M,1st edition, Food Processing Principles and Applications CRCPress

Reference books:

- Manay NS and Shadaksharaswamy M,Latest edition, Food-Facts and Principles, New AgeInternational (P) Ltd. Publishers, New Delhi.
- Sivasankar B. Latest edition. *Food Processing and Preservation*. First Edition. PHILearning.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester I

Course: Major **Scheme of Evaluation(P)**

Subject Title: Practical on Principles of Food Processing and Preservation

Subject Code: FTC152M111 **L-T-P-C: 0-0-6-3** **Total credits: 3**

Course Objectives

Understanding the application of various laboratory equipment's used in food processing and preservation

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** basic food technology laboratory equipment's and Appliances	**BT 1**
CO 2	**Classify** different food groups, determination of moisture content of different food samples	**BT 2**
CO 3	**Construct** the procedures for drying and packaging of fruits and Vegetables	**BT 3**
CO 4	**Analyse** the process of malting from cereals, millets and pulses	**BT 4**

Course Outline -

Modules	Topics (if applicable) & Course Contents	Periods
I.	Introduction to basic laboratory facilities-tools and appliances, basic food processing equipment, care and maintenance of laboratory tools, appliances and equipment.	**20**

II	Identification of foods under different food groups, Determination of moisture content of different food groups like fruits, vegetables, green leafy vegetables, cereals and pulses, Brix analysis by refractometer for different food samples like fruit juices, jam , jelly, Determination of pH of different food products like pickles, fruit juice and sauces	**25**
III	Drying of fruits and vegetables, green leafy vegetables, Fruit/vegetable puree making , Packaging of prepared food products	**20**
IV	Rice flour making from waxy and non-waxy rice, Waxy rice products, Non waxy rice products, Preparation of products from different food groups, Preparation of cereal, millet, pulse malts	**25**
	Total	**90**
	Pedagogy: Lectures, Experiments, Laboratory sessions	

Recommended Texts: As suggested under the theory paper

Semester I

Course: Major **Scheme of Evaluation: (T)**
Subject Title: Basic Food Science
Subject Code: FTC152M102
L-T-P-C: 3-0-0-3 **Total credits: 3**

Course Objectives To train the students with introductory knowledge of food science, scope of food science and its applications in food industries

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**BloomsTaxonomy Level**
CO 1	**Relate** the principle of food science, and food constituents- its properties and functions	**BT 1**
CO 2	**Explain** the basic preservation techniques used in food science	**BT2**
CO 3	**Develop** different types of non-thermal processing techniques used in food Industry	**BT 3**
CO 4	**Analyse** the different advantages and disadvantages of different preservation techniques in terms of increasing shelf Life	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I.	**Introduction:** Introduction to Food Science, Different kinds of Food Industries, Components of Food industries. Scope of food processing and technology. **Food constituents:** Carbohydrates, lipids, proteins, vitamins and minerals, water. Nutritional and chemical properties of food constituents and its function.	**15**
II.	**Introduction to Food preservation techniques:** Pasteurization, Sterilization, Ultra High temperature, Blanching, etc. Low temperature preservation techniques: Cooling, Evaporation, refrigeration and freezing, Drying and their importance in the food processing.	**15**
III.	**Introduction to non-thermal food preservation techniques:** Introduction to new techniques in preservation of food like High Pressure Processing, Ohmic heating, Pulse electric field processing, Irradiation etc.	**15**
IV.	**Basic introduction**: Unit operations in Food Processing, Cleaning, dry cleaning methods, wet cleaning methods, peeling, grading, sorting.	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Reference books:

- Manay NS and Shadaksharaswamy M,Latest edition, Food-Facts and Principles, New AgeInternational (P) Ltd. Publishers, New Delhi.
- Sivasankar B. Latest edition. *Food Processing and Preservation*. First Edition. PHILearning.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester I

Course: SEC 1 **Scheme of Evaluation: (P)**
Subject Title: Fruits and Vegetables Processing **L-T-P-C: 0-0-6-3**
Subject Code: FTC152S111 **Total credits: 3**

Course Objectives

Understanding the physiological parameters for fruits and vegetables processing

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** basic fruits and vegetable varieties and identify their maturity indices	**BT 1**
CO 2	**Classify** the TSS and acidity of different fruits and vegetables	**BT 2**
CO 3	**Construct** the process for identification of spices and additives in different fruits and vegetables	**BT 3**
CO 4	**Analyse** and planning of project for setting up of food processing units	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I.	Identification of various fruit and vegetable varieties. Determination of maturity indices of fruits and vegetables: Days from full bloom (DFFB), firmness. Determination of Starch-iodine ratio, Brix-acid ratio. Dehydration and rehydration of fruits and vegetables	**20**
II	Determination of moisture content of different food products. Determination of TSS of different food products. Determination of acidity of fruits and vegetables	**25**
III	Identification of spices and condiments in fruits and vegetables. Preparation of jam, jelly and marmalade. Preparation of pickles	**30**
IV	Project planning on food processing unit	**15**
	Total	**90**
	Pedagogy: Lectures, Experiments, Laboratory sessions	

Reference books:

- Manay NS and Shadaksharaswamy M,Latest edition, Food-Facts and Principles, New AgeInternational (P) Ltd. Publishers, New Delhi.
- Sivasankar B. Latest edition. *Food Processing and Preservation*. First Edition. PHILearning.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester II

Course: Major **Scheme of Evaluation: (T)**
Level of Course: 100
Title of Paper: Food Microbiology **Subject Code: FTC152M201** **L-T-P-C: 3-0-0-3**
Total credits: 3

Course Objectives
To train the students with various microbial growth control techniques in food preparation.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Relate** the scope, importance, basic techniques of microbiology	**BT 1**
CO 2	**Explain** the various sterilization- both physical and chemical methods	**BT 2**
CO 3	**Develop** the different methods for isolation and preservation of food microbes	**BT 3**
CO 4	**Analyse** the application of food microbiology and its comparison to the industrial fields of microbiology	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I.	**Introduction to Food Microbiology:** Definition, History, phases in microbial growth; Intrinsic (pH, moisture content, oxidation-reduction potential, nutrient content, antimicrobial constituents and biological structure) and Extrinsic (temperature of storage, RH and composition of gas in the environment) Parameters of foods that affect microbial growth.	**15**
II.	**Microbial spoilage of food:** Fruits and vegetables, meat, seafood, cereals, bakery and dairy products. Detection and mechanism of spoilage. HACCP	**15**
III.	**Microbial Control Strategies in Food Preservation**: by Low-Temperature, High Temperature, Drying, Non-thermal Technologies (high pressure processing, pulse electric field, ultrasound, UV radiations, Magnetic energy), high intensity light pulse, hurdle technology, and other combination eg. Manothermosonication.	**15**
IV.	**Probiotics and Prebiotics:** Definition, types, source and Health benefits. Bacterial and yeast probiotics: strains, mechanism and clinical applications. Probiotic foods: curd, yogurt, healthy drinks, etc.	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Textbooks:

- Modern Food Microbiology (4th Ed) by James M. Jay. Chapman &Hall, Inc
- Microbiology by R. P. Singh Kalyani Publishers.

Reference books:

- Fellows PJ, 2016. Food Processing Technology, Principles and Practice. Fourth Edition. Woodhead Publishing
- Sivasankar B. Latest edition. *Food Processing and Preservation*. First Edition. PHI Learning

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester II

Course: Major **Scheme of Evaluation: (P)**
Title of Paper: Practical on Food Microbiology **Subject Code: FTC152M211**
L-T-P-C: 3-0-0-3 **Total credits: 3**

Course Objectives
To train the students with various microbial growth control techniques in food preparation.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Relate** the scope, importance, basic techniques of microbiology	**BT 1**
CO 2	**Explain** the various sterilization- both physical and chemical methods	**BT 2**
CO 3	**Develop** the different methods for identification of food microbes	**BT 3**
CO 4	**Analyse** the application of food microbiology practical on safety implementation	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I.	**Introduction to Microbiology Laboratory:** Familiarization with lab equipment: microscope, autoclave, incubator, Laminar airflow, hot air oven, pH meter. Preparation of nutrient broth, nutrient agar	**15**
II.	**Sterilization Techniques:** Demonstration of physical (autoclaving, hot air oven, filtration) and chemical (alcohol, bleaching powder) sterilization methods	**15**
III.	**Staining Techniques**: Preparation of smear and performance of simple staining, Gram staining, and spore staining.	**15**
IV.	**Measurement of Microbial Growth:** Determination of bacterial growth by turbidimetry (OD measurement) and standard plate count method. Determination of total viable count (TVC) in food samples using standard plate count.	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:
- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Textbooks:

- Modern Food Microbiology (4th Ed) by James M. Jay. Chapman &Hall, Inc
- Microbiology by R. P. Singh Kalyani Publishers.

Reference books:

- Fellows PJ, 2016. Food Processing Technology, Principles and Practice. Fourth Edition. Woodhead Publishing
- Sivasankar B. Latest edition. *Food Processing and Preservation*. First Edition. PHI Learning

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
	70	**30**

Semester II

Course: Major **Scheme of Evaluation: (T)**
Title of Paper: Fermentation Technology **Subject Code: FTC152M202**
L-T-P-C: 3-0-0-3 **Total credits: 3**

Course Objectives
To train the students with various fermentation techniques in food preparation.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic principles, history, and scope of fermentation technology in food processing	**BT 1**
CO 2	**Explain** different types of fermentation processes and fermentor design	**BT 2**
CO 3	**Develop** knowledge of microbial cultures used in food fermentation and their isolation techniques	**BT 3**
CO 4	**Analyse** the manufacturing processes of various fermented food products and their quality control	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I.	Definition, history, and scope of fermentation technology Role of fermentation in food preservation and value addition, Types of fermentation,Microorganisms used in food fermentation, Fermentation media: composition, preparation, and sterilization.	**15**
II.	Basic design and components of a fermentor (bioreactor), Vessel, impeller, sparger, baffles, Temperature control system, pH and dissolved oxygen sensors, Foam control system, Types of fermentors: Batch, fed-batch, and continuous fermentation Airlift fermentor, tower fermentor, Fermentation parameters:Temperature, pH, dissolved oxygen, Agitation and aeration rates, Inoculum development and scale-up	**15**
III.	**Fermented Food Products – I (Dairy and Cereal Based)** – Topics include dairy-based fermented products such as yogurt covering production, starter cultures, and types; cheese including classification, production steps, and ripening; butter and buttermilk; as well as kefir and koumiss. Cereal and legume-based fermented products include bread and sourdough fermentation; idli and dosa batter fermentation; koji (rice fermented with Aspergillus); and soy sauce fermentation.	**15**

IV.	**Fermented Food Products – II (Vegetable, Fruit, and Beverages) –** Topics include vegetable-based fermented products such as sauerkraut (fermented cabbage), kimchi, and pickles (brine fermentation); fruit-based fermented products including vinegar (two-stage fermentation), wine and cider production; fermented beverages such as beer covering raw materials, mashing, fermentation, and conditioning, as well as traditional fermented drinks like handia and rice beer; and quality control in fermented food production covering off-flavors and defects, shelf-life stability, and food safety in fermentation including pathogen control.	**15**
	TOTAL	**48**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Textbooks:

- Modern Food Microbiology (4th Ed) by James M. Jay. Chapman &Hall, Inc
- Microbiology by R. P. Singh Kalyani Publishers.

Reference books:

- Fellows PJ, 2016. Food Processing Technology, Principles and Practice. Fourth Edition. Woodhead Publishing
- Sivasankar B. Latest edition. *Food Processing and Preservation*. First Edition. PHI Learning

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester II

Course: Major **Scheme of Evaluation: (P)**
Title of Paper: Practical on Fermentation Technology **Subject Code: FTC152M212**
L-T-P-C: 3-0-0-3 **Total credits: 3**

Course Objectives
To train the students with various fermentation techniques in food preparation.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic principles, history, and scope of fermentation technology in food processing	**BT 1**
CO 2	**Explain** different types of fermentation processes and fermentor design	**BT 2**
CO 3	**Develop** knowledge of microbial cultures used in food fermentation and their isolation techniques	**BT 3**
CO 4	**Analyse** the manufacturing processes of various fermented food products and their quality control	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I.	**Introduction:** Familiarization with fermentor components, media preparation equipment, and sterilization techniques; preparation of MRS broth for LAB, YPD broth for yeast, and potato dextrose broth for molds.	**15**
II.	**Fermentor and Parameters:** Small-scale batch fermentation using yeast and sugar solution with monitoring of CO_2 production; effect of temperature and pH on fermentation; inoculum development for yogurt/curd fermentation.	**15**
III.	**Dairy and Cereal Fermentation:** Yogurt preparation using *Lactobacillus bulgaricus* and *Streptococcus thermophilus*; paneer preparation; idli/dosa batter fermentation measuring volume increase and pH change; bread dough fermentation; koji preparation using *Aspergillus oryzae*(demonstration)	**15**
IV.	**Vegetable, Fruit, and Beverage Fermentation:** Sauerkraut fermentation monitoring pH and acidity; pickle fermentation; vinegar production via two-stage fermentation; small-scale wine preparation; quality analysis of fermented products (pH, acidity, alcohol, sensory); industrial visit to dairy, brewery, or vinegar unit.	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Textbooks:

- Modern Food Microbiology (4th Ed) by James M. Jay. Chapman &Hall, Inc
- Microbiology by R. P. Singh Kalyani Publishers.

Reference books:

- Fellows PJ, 2016. Food Processing Technology, Principles and Practice. Fourth Edition. Woodhead Publishing
- Sivasankar B. Latest edition. *Food Processing and Preservation*. First Edition. PHI Learning

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester II

Course: SEC II **Scheme of Evaluation: (P)**
Subject Title: Technique in Bakery
Subject Code: FTC152S211
L-T-P-C: 0-0-6-3 **Total credits: 3**

Course Objectives

To provide students with knowledge and practical skills in bakery ingredients, processes, and techniques for producing quality baked products while fostering innovation, safety and entrepreneurial ability.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define the** fundamental baking techniques to prepare standard bakery products**.**	**BT 1**
CO 2	**Classify the** knowledge of ingredient functions and their role in baking.	**BT 2**
CO 3	**Demonstrate the use of** bakery tools and Equipment safely and effectively.	**BT 3**
CO 4	**Analyse** the quality and presentation of baked goods.	**BT 4**

Course Outline

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Fundamentals of Baking**: Basic principles of baking, the history and scope of the bakery industry, and overview of the structure and composition of bakery products. Ingredients such as flour, sugar, fats, milk, eggs, and yeast, explaining their functional roles in dough development and product characteristics. Understanding the chemical and physical changes that occur during baking.	**20**
II	**Bread-Making Techniques**: Principles and techniques involved in bread production. Methods of dough mixing, fermentation, proofing, shaping, and baking. Different types of breads, including white, whole wheat, multigrain, and specialty breads. Common defects in bread and their remedies, along with factors affecting texture, flavor, and shelf life. Practical aspects of bread production in a bakery setting.	**25**
III	**Cakes, Cookies, and Pastries**: Techniques of preparing cakes, cookies, and pastries. Methods like creaming, whisking, and blending methods used in cake preparation, along with the principles of cookie and biscuit tproduction. Identify faults in cakes and pastries.	**30**

IV	**Advanced Bakery Practices and Innovation**: Modern trends, equipment, and innovations in the bakery industry. Nutritional modification of bakery recipes, cost control, hygiene, and safety practices are discussed. **Project Work:** Development and preparation of an innovative baked product with focus on recipe formulation, process standardization, quality evaluation, and cost analysis.	**15**
	Total	**90**
	Pedagogy: Lectures, Experiments, Laboratory sessions	

Recommended Texts: As suggested under the theory paper.

Reference Books:

Bali, P. S. (2016). Theory of bakery and patisserie. Oxford University Press.

Jagarlamudi, L. (2020). Bakery and confectionery products: Processing, quality assessment, packaging and storage techniques. CRC Press, Taylor & Francis Group.

Matz, S. A. (1992). Bakery technology and engineering (3rd ed.). Springer.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
	-70	**30**

Semester III

Course: Major **Scheme of Evaluation: (T)**
Level of Course: 200
Title of Paper: Food Chemistry **Subject Code: FTC152M301**
L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

Understanding the composition of different chemicals present in foods and their relation to its taste and to gain basic knowledge on metabolic roles and functional aspects of food components.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	explain the properties, the composition, structure of nutrients	**BT 2**
CO 2	identify the role of food constituents, nutrients, techniques used for estimation	**BT 3**
CO 3	apply the knowledge of proteins, carbohydrates, fats	**BT 3**
CO 4	analyse the structural and functional properties of lipids, fats and oil, its deficiencies and excess, role of vitamins and minerals and their recommended dietary allowances	**BT 4**

Course Outline:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	Food And Its Constituents: Food and Nutrients - Definition, Classification, and Functions: Role of Water in Food and Human Health, Pigments, Phytonutrients, Antioxidants, Flavour Components.	15
II.	Carbohydrates: Definition, Structure, Properties, Functions, Classification, Dietary Sources, Chemical Reactions Proteins: Definition, Structure, Properties, Functions, Classification, Dietary Sources, Chemical Reactions, Deficiencies and Excess, Recommended Dietary Allowances, chemical reactions	15

III.	Fats and oil: Definition, Structure, Properties, Functions, Classification, Fatty Acid Oxidation, Rancidity of Fats, Emulsions, chemical reactions.Vitamins And Minerals: Classification, Functions, Dietary Sources, Deficiencies and Excess, Recommended Dietary Allowances	15
IV	Enzymes: general characteristics, enzymes in food processing, immobilization of enzymes, Enzymatic and non-enzymatic browning such as Maillard reactions Food Additives and its importance: Need of food additives in food processing and preservation, Characteristics and classification of food additives, major categories of food additives, functions and uses of different types of food additives	15
TOTAL		**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential Learning:

1. Group Discussion
2. Seminar/presentation on any of the relevant topics Case Studies

Texts:

1. Agarwal A and Udipi SA. 2014. Textbook of Human Nutrition. JaypeeBrothers Medical Publishers (P) Ltd.
2. Bamji MS, Krishnaswamy K, and Brahmam GNV. 2009. Textbook of Human Nutrition. Third Edition. Oxford and IBH Publishing Co. Pvt.Ltd.

References:

1. Belitz H.-D, Grosch W, and Schieberle P. 1st edition. Food Chemistry. FourthEdition. Springer.
2. Civille GV and Carr BT. 2016. Sensory Evaluation Techniques. Fifth Edition.CRC Press.
3. Rodwell VW, Bender DA, Botham KM, Kennelly PJ, Weil PA. 2015.Harper's Illustrated Biochemistry. 30th Edition. McGraw Hill Education.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
	-70	**30**

Semester III

Course: Major **Scheme of Evaluation: (T)**
Title of Paper: Cereals, Pulses, Legumes and Oilseed Technology
Subject Code: FTC152M302
L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To understand the technology of milling of various cereals, processing of pulses and oilseeds. To gain knowledge on importance and processing of protein rich products and to introduce concepts of manufacturing alcoholic beverages

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	define the milling techniques for wheat, flour treatments,types of flour etc.	BT 2
CO 2	explain the processing of rice,corn,barley and other cereal crops.	BT 3
CO 3	explain the processing of pulses, milling procedures and antinutritional factors	BT 3
CO 4	apply the knowledge of processing of oilseeds (Soyabean, Coconut), refining of fats and oil, concepts of protein isolates, their sources, properties and uses	BT 4

Course Outline:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	Wheat technology: Wheat-Types, milling, flour grade, flour treatments (bleaching, maturing), flour for various purposes, Products and By- products.	**15**
II.	Rice and other cereal crop technology – Physicochemical properties, milling (mechanical & solvent extraction), parboiling, ageing of rice, utilization of byproducts Corn–Milling (wet & dry), cornflakes, corn flour. Barley- Milling (pear barley, barley flakes & flour) Oats–Milling (oatmeal, oat flour & oat flakes) Sorghum and millets – Traditional & commercial milling (dry & wet)	**15**
III	Pulse technology: Processing- Soaking, Germination, Decortication, Cooking and Fermentation. Changes during germination, Milling- decortication and splitting (dry and wet milling). Antinutritional factors, Factors affecting cooking time	**15**

IV	Oilseeds technology: Sources, Composition, Processing of oil seeds – Soya bean, coconut. Hydrogenation. Refining of fats & oils, bleaching, de odorizing, hydroxylation, Protein isolates, Sources of protein (defatted flour, protein concentrates and isolates), properties and uses, protein texturization, fibre spinning	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Texts:

1. Chakraverty. 2019. Post-Harvest Technology of Cereals, Pulses and Oilseeds, revisedEd., Oxford & IBH Publishing Co. Pvt Ltd.
2. Manay, S. and Sharaswamy, M. 2004. Food Facts and Priniciples. Wiley Eastern Limited

References:

1. Kent, N.L. 2003. Technology of Cereal, 5th Ed. Pergamon Press.
2. Marshall, Rice Science and Technology. 1st edition, Wadsworth Ed., Marcel Dekker, New York

Semester III

Course: Major **Scheme of Evaluation: (P)**
Title of Paper: Practical on Food Chemistry
Subject Code: FTC152M311
L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To gain knowledge about proximate analysis of food samples and their manufacturing techniques Course

Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	find the estimation for total protein, lipid, reducing and non-reducing sugar content in food samples	BT 1
CO 2	demonstrate the ash, moisture, gluten, iodine value determination in food samples	BT 2
CO 3	apply the techniques of dehydration and rehydration of fruits and vegetables and its manufacturing practices	BT 3
CO 4	list preparation methods for fruits and vegetable samples (Jam, jelly, ketchup, pickles)	BT 4

Course Outline

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	1. Introduction to Food Chemistry Laboratory 2. Laboratory Safety and Good Laboratory Practices (GLP) 3. Preparation of Standard Solutions and Reagents 4. Determination of Moisture Content in Food Samples 5. Determination of Total Soluble Solids (TSS) using Refractometer 6. Determination of pH of Food Products	**20**
II.	1. Determination of Ash Content 2. Estimation of Crude Protein Content 3. Estimation of Crude Fat Content 4. Estimation of Crude Fibre Content 5. Calculation of Carbohydrate Content by Difference 6. Proximate Analysis of Different Food Samples	**25**
III.	1. Determination of Vitamin C Content 2. Determination of Total Phenolic Content 3. Determination of Antioxidant Activity 4. Determination of Reducing and Non-reducing Sugars 5. Determination of Pigments (Anthocyanins/Carotenoids)	**20**
IV.	1. Determination of Acid Value 2. Determination of Peroxide Value 3. Determination of Saponification Value 4. Determination of Iodine Value	**25**

	5. Determination of Titratable Acidity	
	TOTAL	**90**
	Pedagogy: Lectures, Experiments, Laboratory sessions	

Texts

1. Nielsen, S. S. (2017). *Food analysis laboratory manual* (3rd ed.). Springer.
2. Ranganna, S. (2010). *Handbook of analysis and quality control for fruit and vegetable products* (2nd ed.). Tata McGraw-Hill Education.

References

1. Nielsen, S. S. (2017). *Food analysis* (5th ed.). Springer.
2. AOAC International. (2023). *Official methods of analysis of AOAC International* (22nd ed.). AOAC International.
3. Belitz, H.-D., Grosch, W., & Schieberle, P. (2009). *Food chemistry* (4th ed.). Springer.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
	-70	**30**

Semester III

Course: IDC

Scheme of Evaluation: (T) **Title of Paper: Basic Food Processing**

Subject Code: FTC152I301

L-T-P-C: 3-0-0-3 **Total credits: 3**

Course Objectives

This course provides an introductory understanding of food processing and preservation technologies for undergraduate students from non-Food Technology disciplines. It familiarizes students with basic food processing operations, preservation methods, packaging, and the processing of common food products. The course aims to develop an understanding of how food processing contributes to food safety, quality, shelf life, and value addition in the food industry.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic concepts, scope and importance of food processing and preservation, and identify major unit operations used in the food industry	**BT 1**
CO 2	**Classify** food processing and preservation methods and explain their applications in improving food safety and shelf life.	**BT 2**
CO 3	**Learn** the principles of food packaging, storage and processing of common food products.	**BT 3**
CO 4	**Analyse** the effects of food processing on quality, nutritional value, safety and shelf life of food products.	**BT 4**

Course Outline

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Unit I: Basic introduction** Definition, scope and importance of food processing technology; concept of food processing and food preservation; need for processing and preservation of foods; Basic food constituents and their relevance in food processing; water activity and food stability; classification of food processing: primary, secondary and tertiary processing; Basic unit operations in food processing, cleaning, sorting, grading, peeling, size reduction, mixing, filtration, distillation and evaporation.	**15**

II	**Unit II: Food Preservation and Processing Technologies** Food spoilage and principles of food preservation. Preservation and processing methods used in the food industry including thermal methods (blanching, pasteurization, sterilization), low-temperature methods (chilling, freezing), water-removal methods (drying, concentration), biological and chemical methods (fermentation, chemical preservatives), radiation-based methods (food irradiation, UV treatment), and emerging technologies (High Pressure Processing, Pulsed Electric Field processing and Ohmic heating).	**15**
III	**Unit III: Food Packaging and Storage** Introduction to food packaging, functions of packaging, common packaging materials, glass, metal, paper and plastics; factors affecting package selection; Modified Atmosphere Packaging (MAP); role of packaging in maintaining food quality and extending shelf life.	**15**
IV	**Unit IV: Processing of common food products and quality evaluation** Processing of cereals, dairy products, fruits and vegetables, bakery products and beverages with selected examples; Effects of processing on food quality, nutritional value, safety and shelf life of foods; Introduction to quality evaluation of processed foods, sensory evaluation; Importance of hygiene and sanitation.	**15**
	Total	**60**
	PEDAGOGY: Lectures, Assignments and Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Texts:

1. RamSaswamy H and Marcotte M,1st edition, Food Processing Principles and Applications CRC Press 18
2. Sivasankar B. Latest edition. *Food Processing and Preservation*. First Edition. PHI Learning.

References:

1. Manay NS and Shadaksharaswamy M, Latest edition, Food-Facts and Principles, New Age International (P) Ltd. Publishers, New Delhi.
2. Potter NH,5th edition, Food Science, CBS Publication, New Delhi.

Semester III

Course: SEC **Scheme of Evaluation: (P)**

Title of Paper: Waste and By-Product Utilization

Subject Code: FTC152S311 L-T-P-C: 0-0-6-3

Total credits: 3

Course Objectives

Understanding the various waste product utilization techniques in food industries for development of new products

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	define the different cereals waste products for product development and waste utilization	**BT 1**
CO 2	classify the different pulses waste products for product development and waste utilization	**BT 2**
CO 3	construct the different fruit waste products for product development and waste utilization	**BT 3**
CO 4	analyse the different vegetables waste products for product development and waste utilization	**BT 4**

Course Outline -

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	1.Utilization of cereal by products for food/ product development/functional ingredients	**25**
II	1.Utilization of pulses by products for food/ product development/functional ingredients	**20**
III	1. Utilization of fruit by product for food/ product development/functional ingredients	**25**
IV	1. Utilization of vegetables by product for food/ product development/functional ingredients	**20**
	Total	**90**
	Pedagogy: Lectures, Experiments, Laboratory sessions	

Recommended Texts: As suggested under the theory papers

Semester IV

Course: Major **Scheme of Evaluation: T**
Title of the Paper: Concepts of Food Engineering **Level 200**
Subject Code: FTC152M401

L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To acquaint students with the fundamentals of food engineering and its process and to understand the basics of designing food plants and systems

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the basic laws of engineering.	**BT 1**
CO 2	**Explain** heat, mass and momentum transfer	**BT 2**
CO 3	**Apply** transport phenomenon in food processing operations	**BT 3**
CO 4	**Solve** problems related to transport phenomenon	**BT 4**

Course Outline

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I	**Physical properties of Foods**: Methods of Estimation of Shape-Roundness, sphericity, roundness ratio, size, volume- platform scale method, density, specific gravity-apparatus, porosity and surface area	**15**
II	**Thermal Properties of Foods**: Definitions-Specific heat, enthalpy, conductivity and diffusivity, surface heat transfer coefficient. Measurement of thermal properties like specific heat, thermal conductivity and thermal diffusivity	**15**
III	**Aerodynamic properties and Frictional properties of Foods**: Aerodynamic Property- definition- drag coefficient, terminal velocity-application in handling and separation of food materials. Frictional property-coefficient of friction, angle of repose, angle of internal friction, application in food handling and storage	**15**
IV	**Rheology and texture of Foods**: Rheology-Rheological classification-viscoelasticity-viscometers, Hookean Body, St Venant body and Newtonian Body. Texture of foods-Methods of textural evaluation, Subjective and objective Method-texture profile method	**15**
	Total	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning

- Group discussion
- Seminar/presentation on any of the relevant topics
- Case studies

Textbooks

- Rao, M. A., Rizvi, S. S. H. and Datta A.K. latest edition. Engineering Properties of Foods: CRC Press.
- Heldman, D. R. latest edition. Food Process Engineering: AVI Publications.
- Toledo, R. T. latest edition. Fundamentals of Food Process Engineering (2 ed.): CBS Publications, New Delhi.

Reference Books

- Warren McCabe, Julian Smith, Peter Harriott. Unit Operations of Chemical Engineering. 7th Edition. McGraw Hill.
- R. Paul Singh, Dennis R. Heldman, and Ferruh Erdogdu. Introduction to Food Engineering.Latest Edition. Elsevier Science.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester IV

Course: Major **Scheme of Evaluation: (T)**
Title of the Paper: Food Product Development
Subject Code: FTC152M402
L-T-P-C: 3-0-0-3 **Total credits: 3**

Course Objectives

To gain knowledge about proximate analysis of food samples and their manufacturing techniques

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	define the Opportunities in the marketplace for new product development, technological advances driving new product development, government's role in new product development	**BT 1**
CO 2	Explain the scope of new product development and modification of traditional methods.	**BT 2**
CO 3	develop the design for production and sensory evaluation.	**BT 3**
CO 4	analyse the development of food products from commercially available ingredients.	**BT 4**

Course Outline

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	New product development: Introduction- new products, customers and consumers, value addition, and market, marketing characteristics of new products-product life cycle and profit picture, opportunities in the marketplace for new product development technological advances driving new product development, government's role in new product development.	**15**
II.	Designing new products: New Food Product Development (NPD) process and activities; recipe development; use of traditional recipe and modification selection of materials/ingredients for specific purposes; modifications for production on large scale, cost-effectiveness, nutritional needs or uniqueness; use of novel food ingredients and novel processing technologies	**15**
III.	Standardization & large-scale production: Process design; Sensory evaluation Food testing lab requirements, different techniques and tests; statistical quality control; comparison of market samples.	**15**
IV	New Food product development using locally available ingredients: Sensory based evaluation of the developed food products and comparison with the commercially available products (as reference products) and its analysis using novel consumer acceptance techniques such as fuzzy logic approaches.	**15**
	TOTAL	**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Texts:

1. Moskowitz, H.R., Saguy,S. and Straus,T. An Integrated Approach to New Food Product Development, CRC Press, 2006 .

References:

1. Gordon W Fuller, "New Food Product Development: From Concept to Marketplace", 3rd Edition, CRC press, Taylor and Francis Group, UK, 2016.
2. Catherine Side., "Food Product Development: Based on Experience", 2nd Edition, Iowa State Press, Blackwell publications, 2008

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester IV

Course: Major **Scheme of Evaluation:** (P)

Title of the Paper: Practical on Concepts of Food Engineering

Subject Code: FTC152M411

L-T-P-C: 0-0-6-3 **Total Credits:** 3

Course Objectives

To provide hands-on experience in the determination of engineering properties of food materials and to familiarize students with the measurement and analysis of transport phenomena relevant to food processing operations.

Course Outcomes

On successful completion of the course the students will be able to:

Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Demonstrate the use of instruments for determination of physical properties of food materials.	BT 1
CO 2	Explain thermal, aerodynamic and frictional properties of food materials through experimentation.	BT 2
CO 3	Apply engineering principles for measurement and analysis of food systems.	BT 3
CO 4	Analyze and interpret engineering data for food process design and operation.	BT 4

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I	**Physical Properties of Foods**1. Determination of dimensions of grains and food materials using Vernier caliper.2. Determination of geometric mean diameter and aspect ratio.3. Determination of sphericity and roundness of food materials.4. Determination of volume using platform scale method and displacement method.5. Determination of bulk density, true density and porosity of food materials.	**25**
II	**Thermal Properties of Foods**1. Determination of specific heat of food materials.2. Determination of thermal conductivity of food materials.3. Determination of thermal diffusivity of food materials.4. Measurement of surface heat transfer coefficient.5. Study of heat transfer through food systems.	**20**
III	**Aerodynamic and Frictional Properties of Foods**1. Determination of terminal velocity of grains and seeds.2. Estimation of drag coefficient of	**20**

	food particles.3. Determination of coefficient of friction on different surfaces.4. Determination of angle of repose.5. Determination of angle of internal friction and its application in storage structures.	
IV	**Image Analysis, Fluid Flow and Rheological Properties**1. Determination of particle size, projected area and perimeter using ImageJ software.2. Colour measurement of food samples using RGB image analysis.3. Verification of Bernoulli's theorem.4. Determination of Reynolds number under different flow conditions.5. Determination of viscosity using viscometer and rheological characterization of food fluids.6. Texture profile analysis of selected food products.	**25**
	Total	**90**

Lectures, Experiments, Laboratory Sessions

Experiential Learning

• Group Discussion
• Seminar/Presentation on any of the relevant topics
• Case Studies
• Data Analysis using ImageJ and Spreadsheet Software

Textbooks

1. Rao, M. A., Rizvi, S. S. H., and Datta, A. K. Latest edition. *Engineering Properties of Foods*. CRC Press.
2. Heldman, D. R. Latest edition. *Food Process Engineering*. AVI Publications.
3. Toledo, R. T. Latest edition. *Fundamentals of Food Process Engineering* (2nd ed.). CBS Publications, New Delhi.

Reference Books

1. Warren McCabe, Julian Smith, and Peter Harriott. *Unit Operations of Chemical Engineering* (7th ed.). McGraw-Hill.
2. Singh, R. P., Heldman, D. R., and Erdogdu, F. Latest edition. *Introduction to Food Engineering*. Elsevier.
3. Mohsenin, N. N. *Physical Properties of Plant and Animal Materials*. Gordon and Breach Science Publishers.

Credit Distribution

Theory	Practical	Experimental Learning
-	90	30

Course: Major **Scheme of Evaluation:** (T)

Title of the Paper: Food Analysis and Instrumentation

Subject Code: FTC152M703

L-T-P-C: 4-0-0-4 **Total Credits:** 4

Course Objectives

To acquaint students with the principles and applications of food analysis techniques and modern analytical instruments used for quality evaluation, safety assessment, and compositional analysis of food materials.

Course Outcomes

On successful completion of the course the students will be able to:

Sl No.	Course Outcome	Blooms Taxonomy Level
CO 1	Understand the principles of food analysis and sampling techniques used in food quality evaluation.	BT 1
CO 2	Explain the principles and applications of classical and instrumental analytical techniques used in food analysis.	BT 2
CO 3	Apply analytical methods for determination of food composition, quality attributes, and contaminants.	BT 3
CO 4	Analyze and interpret analytical data generated through modern food analysis instruments.	BT 4

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I	**Introduction to Food Analysis:** Importance and scope of food analysis; Sampling techniques and sample preparation; Accuracy, precision, sensitivity and validation of analytical methods; Errors in food analysis; Proximate analysis of foods—moisture, ash, protein, fat, crude fibre and carbohydrate determination; AOAC methods and quality assurance in analytical laboratories.	**15**
II	**Spectroscopic Techniques:** Principles, instrumentation and applications of UV-Visible Spectrophotometry, Colorimetry, Fluorimetry, Atomic Absorption Spectroscopy (AAS), Atomic Emission Spectroscopy (AES), Fourier Transform Infrared Spectroscopy (FTIR), Near Infrared Spectroscopy (NIR) in food analysis.	**15**
III	**Chromatographic Techniques:** Principles, instrumentation and applications of Paper Chromatography, Thin Layer Chromatography	**15**

	(TLC), Gas Chromatography (GC), High Performance Liquid Chromatography (HPLC), Ion Exchange Chromatography and Gel Filtration Chromatography; Applications in analysis of food additives, pigments, vitamins and contaminants.	
IV	**Advanced Analytical Instruments:** Principles and applications of Mass Spectrometry (MS), Gas Chromatography-Mass Spectrometry (GC-MS), Liquid Chromatography-Mass Spectrometry (LC-MS), Differential Scanning Calorimetry (DSC), Texture Analyzer, Colorimeter and Electronic Nose; Instrumentation for food quality, safety and shelf-life evaluation; Data acquisition, interpretation and reporting of analytical results.	**15**
	Total	**60**

Pedagogy

Lectures, Assignments, Seminars

Textbooks

1. Nielsen, S. S. Latest edition. *Food Analysis*. Springer.
2. Pomeranz, Y., and Meloan, C. E. Latest edition. *Food Analysis: Theory and Practice*. Springer.
3. Nollet, L. M. L. Latest edition. *Handbook of Food Analysis*. CRC Press.

Reference Books

1. Nielsen, S. S. Latest edition. *Food Analysis Laboratory Manual*. Springer.
2. Nollet, L. M. L., and Toldrá, F. Latest edition. *Handbook of Food Analysis Instruments*. CRC Press.
3. Skoog, D. A., Holler, F. J., and Crouch, S. R. Latest edition. *Principles of Instrumental Analysis*. Cengage Learning.
4. Christian, G. D. Latest edition. *Analytical Chemistry*. Wiley.

Credit Distribution

Theory	Practical	Experimental Learning
60	-	30

Course: Major **Scheme of Evaluation:** (P)

Level of Course: 400

Title of the Paper: Practical on Food Product Development

Subject Code: FTC152M713

L-T-P-C: 0-0-6-3 **Total Credits:** 3

Course Objectives

To provide practical knowledge on food product development, standardization, sensory evaluation, packaging, and commercialization of value-added food products using locally available food resources.

Course Outcomes

On successful completion of the course the students will be able to:

Sl No.	Course Outcome	Blooms Taxonomy Level
CO 1	Demonstrate the basic principles involved in food product development and standardization.	BT 1
CO 2	Explain the role of ingredients, processing methods, and sensory attributes in product development.	BT 2
CO 3	Develop value-added food products using locally available food resources.	BT 3
CO 4	Analyze the quality, acceptability, packaging, and market potential of developed food products.	BT 4

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I	**Product Formulation and Standardization** Selection of raw materials and ingredients for product development. Standardization of formulations for cereal, pulse, fruit and vegetable-based products. Development of low-cost and nutrient-enriched food products. Product optimization using ingredient variation.	**20**
II	**Development of Value-Added Food Products**1. Development of bakery products using functional ingredients.2. Development of fruit and vegetable-based value-added products.3. Development of convenience	**25**

	and ready-to-eat food products.4. Development of traditional food products with nutritional modification.	
III	**Quality Evaluation and Sensory Analysis**1. Sensory evaluation using hedonic rating scale.2. Difference and preference tests for food products.3. Evaluation of physical and chemical quality attributes.4. Shelf-life assessment and consumer acceptability studies.	**20**
IV	**Packaging, Cost Analysis and Product Commercialization** 1. Selection of packaging materials for developed products.2. Label design and nutritional labelling.3. Product costing and economic feasibility analysis.4. Preparation of project report and market survey for developed food products.5. Presentation and demonstration of developed products.	**25**
Total		**90**

Pedagogy

Lectures, Experiments, Laboratory Sessions

Textbooks

• Moskowitz, H. R., Saguy, S., and Straus, T. Latest edition. *An Integrated Approach to New Food Product Development*. CRC Press.
• Fuller, G. W. Latest edition. *New Food Product Development: From Concept to Marketplace*. CRC Press.
• Earle, M., Earle, R., and Anderson, A. Latest edition. *Food Product Development: Maximizing Success*. Woodhead Publishing.

Reference Books

• Side, C. Latest edition. *Food Product Development: Based on Experience*. Blackwell Publishing.
• Brody, A. L. Latest edition. *Developing New Food Products for a Changing Marketplace*. CRC Press.
• MacFie, H. Latest edition. *Consumer-Led Food Product Development*. Woodhead Publishing.

Credit Distribution

Theory	Practical	Experimental Learning
-	90	30

Course: Major **Semester V**

Scheme of Evaluation:(T)

Title of the Paper: Dairy Technology **Subject Code: FTC152M501**

L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To acquaint students with production and processing technologies for product development and value addition of various dairy based products.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** the basic composition of milk, physical and chemical properties and various grades of milk. .	**BT 1**
CO 2	**Explain** the processing techniques of marketed milk, pasteurization methods of milk.	**BT 2**
CO 3	**develop** the ability to understand various types of milk and their properties.	**BT 3**
CO 4	**analyse t**he manufacturing methods of fermented and indigenous milk products.	**BT 4**

Modules	**Topics (if applicable) & Course Contents**	**Period s**
I.	**Introduction:** Milk - Definition, sources, and composition of milk, factors effecting composition of milk, physiochemical properties of milk, grading of milk-definition and types of grades, collection and transportation of milk.	**15**
II.	**Processing of market milk:** Flowchart of milk processing, Reception Different types of cooling systems. Clarification and filtration process standardization-Pearson's square method, pasteurization-LTLT, HTST and UHT process-continuous pasteurizer, Sterilization and Homogenization Cream separation-centrifugal cream separator, bactofugation	**15**
III.	**Special Milk:** Skim milk, evaporated milk, condensed milk, standardized milk, toned milk, double toned milk, flavoured milk, reconstituted milk.	**15**

IV	**Indigenous and Fermented milk products: Product** description, methods for manufacture of butter, cheese, ice cream, khoa, channa, paneer shrikhand ghee. Spray drying system: dried milk- whole milk and skim milk powder Instantization of milk.	**15**
	TOTAL	**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Text Books:

- Dairy Technology: Principles of Milk Properties and Processing, P. Walstra, T.J. Geurts, A. Noomen, and J.S. Van Boekel, Marcel Dekker, Illustrated Edition, 1999.
- Outlines of Dairy Technology, Sukumar De, Oxford University Press, 3rd Edition, 2006
- Food Engineering and Dairy Technology, HG Kessler, Verlag- A Kessler publication, 1981.

Reference Books:

- Dairy processing & Quality assurance, Chandan RC, Kilara A & Shah NP, Blackwell publishing, 2008
- Milk processing and Quality management, Tamime AY, Blackwell publishing, 2009

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester V

Course: Major **Scheme of Evaluation:(T)**

Title of the Paper: Animal Product Technology

Subject Code: FTC152M502

L-T-P-C: 4-1-0-4 **Total credits: 4**

Course Objectives

To acquaint students with production and processing technologies for product development and value addition of various animal and egg products.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** the basic composition and classification of poultry meat. .	**BT 1**
CO 2	**Explain** the slaughtering methods of various categories of animals.	**BT 2**
CO 3	**develop** the ability to understand the classification of various grades of animals	**BT 3**
CO 4	**analyse t**he processing methods of eggs and egg products.	**BT 4**

Modules	**Topics (if applicable) &Course Contents**	**Periods**
I.	**Introduction**: Classification, composition and nutritional value of poultry meat, Effects of feed breed and stress on production of meat animals and thei quality, Meat Quality-color, flavor, texture, Water-Holding Capacity (WHC) Emulsification capacity of meat	**15**
II.	**Slaughtering of animals:** buffalo, sheep/ goat, poultry, pig; Antemortem examination of meat animals, post-mortem examination of meat, Grading, Post-mortem changes of meat; Importance of by-products utilization, classification and uses of by-products Refrigeration and freezing, thermal processing- canning of meat, retort pouch, dehydration, irradiation, meat curing, Sausages-processing, types and defects, Packaging of meat	**15**

III.	**Classification of freshwater and marine fish**: composition, storage and transport, Preservation: Drying, smoking, curing, freezing, salting and Canning, Surimi-Introduction, fish muscle proteins, the surimi process comparison of surimi and fish mince products, Fish protein concentrates (FPC), fish protein extracts (FPE), fish protein hydrolysis (FPH); Flowchart o Indigenous products- Fish sauce and Paste	**15**
IV	**Structure, composition and nutritional value of egg**: egg processing byproducts, egg powder manufacture, egg coatings, Factors affecting egg quality and measures of egg quality, processing of lecithin	**15**
	TOTAL	**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential

Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Text Books:

- Stadelman, W. J., Newkirk, D., & Newby, L. (2002). Egg science and technology. 4th ed. New Delhi: CBS Publication.
- Pearson, A. M., & Gillett, T. A. (1997). Processed meats. 3rd ed.New Delhi: CBS Publication

Reference Books:

- Shai, Barbut. (2005). Poultry Products Processing. CRC Press
- Parkhurst, C., &Mountney, G. J. latest edition,. Poultry meat and egg production. New Delhi: CBS Publishers.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Textbooks

- Pearson, A. M., & Gillett, T. A. (1997). Processed meats. 3rd ed.New Delhi: CBS Publication.
- Sen, D.P. (2005).Advances in Fish Processing Technology. Allied Publishers Pvt.Limited.
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Reference Books:

- Shai, Barbut. (2005). Poultry Products Processing. CRC Press
- Parkhurst, C., &Mountney, G. J. latest edition,. Poultry meat and egg production. New Delhi: CBS Publishers.

Semester V

Course: Major **Scheme of Evaluation: (T)**

Title of the Paper: Food Quality and Sensory Evaluation

Subject Code: FTC152M503 **L-T-P-C: 4-0-0-4** **Total credits: 4**

Course Objectives

To understand knowledge of quality control and total quality management system in food industry and knowledge about food hygiene and importance of safe food and Food safety management system.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the concepts of food quality and role of total quality management system in food industry	**BT 1**
CO 2	**understand** the quality of food products using various techniques	**BT 2**
CO 3	**apply** the national & international food laws and regulations for quality of foods	**BT 3**
CO4	**explain** the sensory analysis techniques.	**BT 4**

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Quality and Assurance:** Definition, scope, importance and difference, Total quality control and (TQC) Total quality management (TQM), Statistical quality control. Definition, importance, scope and difference between food quality and food safety.	**15**
II.	**Raw materials & Finished product quality**: Quality parameters and evaluation procedures: appearance, color, texture, viscosity, consistency, flavour etc. Food standards and laws: International – Concept of Codex Alimentarius, HACCP, GMP, GHP, USFDA, ISO 9000, ISO 22000, ISO 14000. Export Quality Control and Inspection act (1963), Environment Protection Act (1986), WTO & GATT, etc. GMP, GHP. National – Compulsory and voluntary trade and Company standards. Consumer Protection Act (1986), BIS/IS, Food Safety and standards – 2006, FPO, MPO, MMPO, Agmark. GMP, GHP.	**15**

III	**Quality Certification & Accrediation**: Introduction and procedure Prevention o food adulteration Act: Food Adulteration: definition, common adulterants in different foods, contamination, method of detection, Food additives and legislation PFA specification for food products, Nutritional labeling Risk and Hazard associated with Food: Food hazards, sources of hazard, classification, Food safety; prevention and control, Statistical quality control. HACCP, Quality costs.	**15**
IV	**Sensory Evaluation:** Selection of panel of judges, Prerequisite for sensory analysis, application of consumer tests; control of factors affecting of sensory verdict, Instrumental measurements of sensory attribute of foods sensory characteristics of foods, types of tests, Texture profile analysis. Correlation between instrumental and Sensory analysis of food quality attributes.	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics

Textbooks:

1. Food Quality Assurance: Principles and Practices, I. Ali, CRC Press, 2nd Edition, 2004.
2. Quality Assurance in Food Industry: a Practical Approach, J. A. Vasconcellos, CRC Press, 1st Edition, 2003.
3. Statistical Quality Control for the Food Industry, M. R. Hubbard, Kluwer Academic/ Plenum Publishers, 3rd Edition, 2003.

References:

1. HACCP in Meat, Poultry and Fish Processing, A. M. Pearson and T.R. Dutson, Kluwer Academic Publishers, 2nd Edition, 1999.
2. Food Safety Contaminants and Toxins, J. P. F. D'Mello, Oxford University Press, 1st Edition, 2003.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester V

Course: Major **Scheme of Evaluation:(T)**

Title of the Paper: Practical on Dairy and Animal Product Technology

Subject Code: FTC152M511 **L-T-P-C: 0-0-8-4** **Total credits: 4**

Course Objectives

To acquaint students with production and processing technologies for product development and value addition of various animal and egg products.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** the basic laboratory equipment's and their handling practices .	**BT 1**
CO 2	**explain** the methods for estimation of moisture and protein content in meat.	**BT 2**
CO 3	**develop** the methods to study the shelf life of eggs and evaluate the quality of eggs.	**BT 3**
CO 4	**analyse** the evaluation methods for various fish categories.	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I.	Introduction to the Basic Laboratory Practices and Equipment Functioning and use of compound microscope: Cleaning and sterilization of glassware Preparation and sterilization of nutrient broth Cultivation and sub-culturing of microbes Preparation of slant, stab and plates using nutrient agar	20
II.	Estimation of moisture content of meat, Cut out analysis of canned meats/retort pouches Estimation of protein content of meat. Analysis of frozen meat/meat emulsion products	25
III.	To study shelf-life of eggs by different methods of preservation. Evaluation of eggs for quality parameters (market eggs, branded eggs). To perform freezing of yolk/albumen Meat/Egg product formulation.	20
IV	Quality evaluation of fish/prawn. Subjective evaluation of Fresh Fish. Cut out examination of canned fish. (Sardine, Mackerel, Tuna) Fish/Milk product formulation	25
	TOTAL	90
Pedagogy: Lectures, Assignments, Seminars		

Level: Semester VI

Course: Major **Scheme of Evaluation:(T)**

Title of the Paper: Food Toxicology **Subject Code: FTC152M601**

L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To acquaint students with. contaminants and toxins associated with food processing.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** the basic terms of food toxicology and classifications. .	**BT 1**
CO 2	**explain** the concepts of Carcinogenesis, Mutagenesis, Teratogenesis	**BT 2**
CO 3	**develop** the knowledge to identify the naturally occurring toxicants & food contaminants	**BT 3**
CO 4	**analyse** the different types of food additives as toxicants.	**BT 4**

Course Outline

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	Food Toxicology: Introduction. Historical evolution. Bibliography. Toxicological concepts. Intoxication classification. Classification of toxic substances.	**15**
II.	Carcinogenesis, Mutagenesis, Teratogenesis of Organ, Agricultural and industrial contaminants in foods: pesticides residues in fruits and vegetables, metal contaminants in foods and their toxicity in human body; animal drug residues in food and water, dioxins and related compounds in food; metals such as lead, arsenic and mercury	**15**

III.	Naturally occurring toxicants & food contaminants: Sea food toxins, biogenic amines, mutagens & carcinogens in heated & processed foods, coffee & methylxanthines, toxicity of mushrooms alkaloids compounds, glucosinolates, protease inhibitors, phytate.	**15**
IV	Food additives as toxicants: Sweeteners; toxicants formed during food processing such as Maillard reaction products acrylamide, benzene; risk of genetically modified food, food supplements, persistent organic pollutants	**15**
	TOTAL	**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Text Books:

1. Stadelman, W. J., Newkirk, D., & Newby, L. (2002). Egg science and technology. 4th ed. New Delhi: CBS Publication.
2. Pearson, A. M., & Gillett, T. A. (1997). Processed meats. 3rd ed.New Delhi: CBS Publication.
3. Sen, D.P. (2005). Advances in Fish Processing Technology. Allied Publishers Pvt.Limited.

Reference Books:

- Billoneyer, F. W. 1962. Text book of polymer science, Interscience. London.
- Abdulla, M., Vohora, S. B and Athar, M. 1995. Trace and toxic elements

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester VI

Course: Major

Scheme of Evaluation:(T)

Title of the Paper: Food Safety Standard and Quality Control

Subject Code: FTC152M602 **L-T-P-C: 4-0-0-4** **Total credits: 4**

Course Objectives

To Understand the role of various safety measurements to be taken in food industries.

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Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**BloomsTaxonomy Level**
CO 1	**define** the types of hazards and factors affecting food safety	**BT 1**
CO 2	**Explain** the different types of food borne pathogens	**BT 2**
CO 3	**develop** the knowledge for food certification and prerequisites	**BT 3**
CO 4	**analyse** the quality concepts and quality attributes	**BT 4**

Course Outline

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Definition of food safety**: Types and examples of hazards: biological, chemical, physical hazards, Factors affecting Food Safety, Importance of Safe Foods. Food hazards, Impact of food safety on health. Control measures.	**15**
II.	**Hazards of biological origin:** Indicator organisms, Food borne pathogens: Bacteria, Viruses, Eukaryotes, Mycotoxins, Sea food and shell fish poisoning. Hazard management: Control parameters, Temperature control, food storage,	**15**

	Product design., Microbiological Assessment and categories of food based on microbial quality, Basic steps in detection of food borne pathogens, Water analysis.	
III.	**Tools of food safety management**: Food certification, Prerequisites- GHPs, GMPs, SSOPs, HACCP, ISO series, TQM - concept and need for quality, components of TQM, Kaizen. Risk Analysis, Accreditation and Auditing, Definition – Methods to detect adulterant of various foods, Objective of Food Laws, Major Food Laws and Regulations of India and Regulation of Food Sanitation, Idea of PFA, FPO, MPO, Agmark, BIS, FSSAI, GMO	**15**
IV	**Introduction to food quality management** – Definition, quality concepts, quality, quality perception, quality attributes, safety, health, sensory, shelf life, convenience, extrinsic attributes, factors affecting food behaviours and quality.	**15**
	TOTAL	**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Text Books:

1. 1. Pieternel A, Luning, Willem J. Marcelis, Food Quality Management Technological and Managerial principles and practices, Wageningen, 2009.
2. Lawley, R., Curtis L. and Davis,J. The Food Safety Hazard Guidebook, RSC publishing, 2004

Reference Books:

1. Pieternel A, Luning, Willem J. Marcelis, Food Quality Management Technological and Managerial principles and practices, Wageningen, 2009.
2. Lawley, R., Curtis L. and Davis,J. The Food Safety Hazard Guidebook, RSC publishing, 2004

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Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester VI

Course: Major **Scheme of Evaluation:(T)**

Title of the Paper: Unit Operations in Food Processing

Subject Code: FTC152M603

L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To acquaint students with knowledge of types of operation techniques used in food processing industry.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** the types of screens and their principles.	**BT 1**
CO 2	**Explain** the types of separators and their operation techniques.	**BT 2**
CO 3	**develop** the knowledge to identify the size reduction methods	**BT 3**
CO 4	**analyse t**he different techniques of drying.	**BT 4**

Course Outline

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	Screening; types of screens; Grizzly; Revolving screen; Shaking screen; Rotary screen; Vibratory screen; Horizontal screen; Perforated metal screens; wire mesh screen; Ideal and Actual screen; Effectiveness of screen; Air screen cleaners;	**15**
II.	Definition and Introduction to Separation; Types of Separators- Disk Indented cylinder, Spiral, Specific Gravity, Destoner, Inclined Draper, Velve roll, Pneumatic & aspirator, separation-based fluidization technique, Magnetic and Cyclone Separator.	**15**
III.	Size reduction procedures- Crushing, Impact, Shearing, cutting; Cerea grinding Degree of grinding; Size reduction machinery- crusher, grinder attrition mills hammer mill, ball mills, rietz mill & oil expression and extractions- hydraulic press screw press.	**15**

IV	Utilities of Drying; thermal properties; Equilibrium moisture content, Drying theories; methods of drying; - Contact drying, Convective drying freeze drying, radiation drying, Superheated steam, Drying rate period; types of dryers-Deep bed, Flat bed, Continuous, Recirculating, LSU, Fluidized bed, Rotary, Tray, Tunnel and solar, etc.	**15**

	TOTAL	**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics
- Case Studies

Text Books:

1. 1.Sahay KM & Singh KK 1994. Unit Operation of Agricultural Processing. Vikash Publication House.
2. 2.Potter NN & Hochkiss 1997 Food Science 5th Ed. CBS

Reference Books:

- Potty VH & Mulky MJ 1993 Food Processing. Oxford & IBH. 5. Ramaswamy H & Marcotte M. 2006 Food Processing: Principles & Applications. Taylor & Francis.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester VI

Course: Major **Scheme of Evaluation: (T)**

Title of the Paper: Nutraceuticals and Functional Foods

Subject Code: FTC152M604 **L-T-P-C: 4-0-0-4**

Total credits: 4

Course Objectives

To understand the Fundamental knowledge of nutraceuticals and functional foods and knowledge about health promoting effect of nutraceuticals and various metabolic disorders and their prevention and treatment by nutraceuticals and processing and storage of functional foods.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms TaxonomyLevel**
CO 1	**define** knowledge of nutraceuticals and functional foods	**BT 1**
CO 2	**explain** the health promoting effect of nutraceuticals.	**BT 2**
CO 3	**explain** role of functional beverages and herbs and their market aspects	**BT 3**
CO 4	**apply** the role of functional beverages and herbs and their market aspects	**BT 4**

Course Outline:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Technological Aspects of Nutraceuticals and functional Foods:** Defining nutraceuticals and functional foods, nature, type and scope. Nutraceuticals and functional foods applications and their health benefits, classification based on chemical and biochemical nature with suitable and relevant descriptions.	**15**

II.	**Nutraceuticals for Specific Diseases:** Food recommended and restricted in metabolic disorders and disturbances, gastrointestinal disorders; fever and infection; liver, gall bladder and pancreatic disturbances; blood, circulatory and cardiac diseases; urinary and musculoskeletal diseases; allergies.	**15**

III	**Functional Role of Food Components:** Antioxidants, phytochemicals, isoflavones, lycopene, their role in Nutraceutical and functional foods, dietary fibers and complex carbohydrates as functional food ingredients. Proteins as a functional food ingredient, probiotic foods and their functional role.	**15**
IV	**Role of Specific Food Products as a Functional Food** Herbs as functional foods health promoting activity of common herbs. Cereal products as functional foods Oats, Wheat bran, rice bran etc. Functional vegetable products, oil seeds and sea foods. Coffee, tea and other beverages as functional foods/ drinks and their protective effects	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics

Texts

:

1. Handbook of Nutraceuticals Pathak YV, CRC Press, 2009
2. Innovations in Healthy and Functional Foods Ghosh D et al., CRC Press,2012

References:

1. Handbook of Nutraceutical and Functional Foods Wildman REC
2. Anti-angiogenic Functional and Medicinal Foods Losso JN

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Semester VI

Course: Major **Scheme of Evaluation: (P)**

Title of Paper : Practical on Food Safety and Standard

Subject Code: FTC152M611 **L-T-P-C: 0-0-8-4** **Total credits: 4**

Course Objectives

understanding of food plant sanitation, traditional food processing, and fermentation techniques and applying hygiene practices, conducting microbial analyses, and ensuring quality control in food production

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** key concepts related to food plant sanitation, traditional food processing, and fermentation technologies, including hygiene and safety standards	**BT 1**
CO 2	**classify** various traditional food products and fermentation methods based on their preparation techniques, ingredients, and regional variations	**BT 2**
CO 3	**construct** practical solutions for enhancing food safety and quality during the production of traditional and fermented foods	**BT 3**
CO 4	**analyse** the impact of processing techniques on the microbial quality, safety, and nutritional value of traditional and fermented food products	**BT 4**

Course Outline –

Modules	Topics (if applicable) & Course Contents	Periods
I.	1. Introduction to basic laboratory facilities-tools and appliances 2. Introduction to basic food processing layout design 3. Cleaning and maintenance of the food processing equipment	**20**
II	1. National and international food safety regulations 2. Food Safety and Standards Authority of India (FSSAI): objectives, functions, and regulations 3. Codex standards and food safety guidelines 4. Food standards, labeling requirements, and food quality assurance 5. Food inspection and certification systems	**25**
III	1. Hazard Analysis and Critical Control Point (HACCP) 2. Sanitation Standard Operating Procedures (SSOP) 3. Quality management systems in food industries 4. Food safety monitoring and documentation 5. Traceability and recall systems	**20**
IV	1.Safety aspects of traditional food processing 2.Microbial quality of fermented foods 3.Fermentation and food preservation 4. Foodborne pathogens and spoilage microorganisms 5. Quality evaluation of traditional and fermented food products 6. Strategies for improving safety, shelf life, and nutritional quality of traditional and fermented foods	**25**
	Total	**90**
	Pedagogy: Lectures, Experiments, Laboratory sessions	

Recommended Texts: As suggested under the theory papers

Level: Semester VII

Course:

Title of the Paper: Research Methodology

Subject Code: FTC152M701

L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To understand design a research problem by applying appropriate experimental design and statistically analyze the collected data and explain the ethics in research and plagiarism and write research papers, reports and research proposals

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** research problem by applying appropriate experimental design	**BT 1**
CO 2	**explain** statistically analyze the collected data.	**BT 2**
CO 3	**explain** the ethics in research and plagiarism	**BT 3**
CO 4	**apply to** write research papers, reports and research proposals	**BT 4**

Course Outline:

Modules	Topics (if applicable) & Course Contents	Periods
I.	**Testing of hypothesis**: Introduction - Large sample tests based on normal distribution - Test for single mean, difference between means - proportion, difference between proportions - standard deviation, difference between standard deviation -Chi-square test for goodness of fit - Independence of attributes.	**15**
II.	**Analysis of variance:** Small sample tests based on t and F distribution-Test for, single mean, difference between means, Paired t-test, test for equality of variances. ANOVA one -way classification, Two-way classification.	**15**
III	**Design of Experiments:** Objectives, strategies, Experimental design, Factorial design, Central composite design, Fractional factorial design, Taguchi's approach to design of experiments. Model development: Empirical model development, Validity of predicted equation, Graphic analysis, Adjusted coefficient of determination, F test for lack of fit. Statistical quality control: Introduction-Process control-control charts for variables-X and R, X and s charts control charts for attributes: p chart, np chart, c chart	**15**
IV	**Research:** Types, Research process and steps in it, Hypothesis, Research proposals and aspects; Research Design: Need, Problem Definition, variables, research design concepts, Literature survey and review, Research design process, Errors in research; Research Modeling: Types of Models, Model building and stages, Data consideration and testing, Heuristic and Simulation modeling. Report Writing: Pre writing considerations, Thesis writing, Formats of report writing and publications in Research journals. Ethical issues, ethical committees Commercialization, Copy right, royalty, Intellectual property rights and patent law, Trade Related aspects of Intellectual Property Rights, Plagiarism, Citation and acknowledgement, Reproducibility and accountability**.**	**15**

	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics

Texts:

1. Montgomery, D. C. Design and Analysis of Experiments, Wiley India, 2007.
2. Montgomery, D.C. and Runger, G.C. Applied Statistics & Probability for Engineers, Wiley India, 2007.

References:

1. Kothari, C.K. Research Methodology Methods and Techniques, New Age International, 2004.
2. Gupta S.C. & Kapoor V.K., "Fundamentals of Mathematical Statistics", Sultan Chand and Sons, New Delhi, 11th edition, 2007.
3. Ewans W. & Grant G., "Statistical Methods in Bio informatics – An Introduction", Springer, 2nd edition, 2005.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Level: Semester VII

Course: Major **Scheme of Evaluation:**

Title of the Paper: Fermentation Technology

Subject Code: FTC152M703

L-T-P-C: 3-1-0-4 **Total credits: 4**

Course Objectives

Understanding food processing through fermentation, including microbial biochemistry, fermentation techniques, equipment used, and production methods for various fermented foods

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**explain** the fundamental principles of fermentation, including the role of microorganisms as starter culture and food material as a substrate	**BT 2**
CO 2	**identify** the role of different microorganism, food materials and environmental conditions in producing fermented food products	**BT 3**
CO 3	**apply** the principle of microbiology and food processing to understand the production of various types fermented food products	**BT 3**
CO 4	**analyse** the starter cultures, raw materials, processing parameters to ensure the safe production of high-quality fermented food products	**BT 4**

Course Outline:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Introduction to fermentation:** History of fermented foods, traditional and modern fermented foods, properties of fermented foods, Micro-organisms in fermented foods (Lactic acid bacteria, Acetobacter, Gluconobacter, Yeast and Molds etc.), Biochemical pathways of metabolic reactions for utilization of carbon sources and formation of different metabolites by microorganisms, types of fermentation, Starter cultures (bacterial starter culture, yeast starter culture and fungal starter culture), Media formulation	**15**
II.	**Fermenters** - Gaden's Fermentation Classification, Design and Operation of Fermenters, Types of Reactors (Packed Bed Reactor, Fluidized Bed Reactor, Trickle Bed Reactor, Bubble Column Reactor), Scale-Up of Bioreactors, Fermenter Design (Body Construction and Materials, Temperature Control Systems, Aeration and Agitation Systems, Stirrer	**15**

	Glands and Bearings, Baffles, Valves and Steam Traps, Pressure Control Valves).	

III.	**Downstream Processing:** Equipment for recovering fermented products; microfiltration and ultrafiltration systems for cell and fermentation medium separation and concentration of product-containing media; chromatographic systems for separation; solvent extraction techniques for product recovery; methods for evaporation and crystallization; centrifugation processes, including different types of centrifuges; drying methods	**15**
IV	**Fermentation Processes and Product Quality Control**: Preparation of Fermented Dairy Products (Yogurt and Shrikhand), Traditional Indian Fermented Foods (Idli, Dosa, Dhokla), Cheese Production Processes; Alcoholic Beverages (Wines, Whiskies, Beers, Rum) Production Process- Malting (steeping, germination, kilning, curing), separation of wort and wort boiling with hops addition, Fermentation, separation, maturation, carbonation, and packaging; Monitoring Fermentation Parameters (pH, Temperature, Time), Quality Assurance Techniques for Fermented Products	**15**
TOTAL		**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential Learning:

1. Group Discussion
2. Seminar/presentation on any of the relevant topics

Case Studies

Texts:

1. Hutkins, R. W. (2018). Microbiology and Technology of Fermented Foods.
2. Mansi EMTEL, Bryce CFA. Fermentation Microbiology and Biotechnology, 2nd Edition, Taylor & Francis Ltd, UK, 2007

References:

1. Principles of Fermentation Technology, P.F. Stanbary, A. Whitaker, Hall, 3rd Edition, Elsevier Publ. 2016
2. Tamang, J. P. (2010). Fermented Foods and Beverages of the World.
3. Ray, B., & Joshi, V. K. (2014). Fermented Foods: Part I—Biochemistry and Microbiology.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Level: Semester VII

Course: Major

Scheme of Evaluation: (P)

Title of Paper: Practical on Research Methodology and Fermentation Technology

Subject Code: FTC152M711

L-T-P-C: 0-0-8-4 **Total credits: 4**

Course Objectives

Understanding of food plant sanitation, traditional food processing, and fermentation techniques and applying hygiene practices, conducting microbial analyses, and ensuring quality control in food production

Course Outcomes

On successful completion of the course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** the basics of using tools like Excel and the statistical analytical tool	**BT 1**
CO 2	**Explain** the use of the Research tool and how to use it, and the test of significance, post hoc analysis	**BT 2**
CO 3	**construct** practical solutions for enhancing food safety and quality during the production of traditional and fermented foods	**BT 3**
CO 4	**analyse** the impact of processing techniques on the microbial quality, safety, and nutritional value of traditional and fermented food products	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I.	1. Experiment 1: Introduction to MS Excel for Research Data Management Familiarize students with Excel interface, Data entry, formatting, sorting, and filtering Experiment 2: Data Visualization Using ExcelCreate and interpret different charts:Bar Chart, Column Chart, Pie Chart, Line Graph, Scatter Plot, Histogram Experiment 3: Descriptive Statistical Analysis Using Excel	**20**

	Calculate: Mean, Median, Mode, Standard Deviation, Variance, Range, Coefficient of Variation	
II	Experiment 4: Linear Regression and Correlation Analysis in Excel Scatter plot generation, Calculation of Correlation coefficient, Coefficient of determination (R^2), Linear regression model fitting, Interpretation of regression equation	**25**
III	Experiment 5: Prepare curd/yogurt through natural lactic acid fermentation and measure acidity, microbial count, and texture Experiment 6: Performing bread fermentation using baker's yeast Experiment 7: Preparation of Gulab jamun using traditional methods, focusing on the frying and sugar syrup processes	**20**
IV	Experiment 8: Implementation of hygienic protocols during the preparation of fermented foods and assess the reduction in microbial contamination Experiment 9: Shelf-life analysis on traditionally processed and fermented foods under controlled and open environments Experiment 10: Preparation of a starter culture for alcoholic beverage production Preparation of acetic acid (vinegar) by fermentation of fruit juice	**25**
	Total	**90**
	Pedagogy: Lectures, Experiments, Laboratory sessions	

Recommended Texts: As suggested under the theory papers

Level: Semester VII

Course: Minor **Scheme of Evaluation: (T)**

Title of the Paper: Food Business Management

Subject Code: FTC152N701

L-T-P-D : 3-1-0-4 **Total credits: 4**

Course Objectives

Understanding of the fundamental principles and practices necessary for effectively managing and operating food businesses, with an emphasis on marketing, operations, finance, and entrepreneurship within the food industry

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**explain** the key concepts and principles of food business management, including operational, marketing, and financial strategies	**BT 2**
CO 2	**identify** the various sectors within the food industry and the specific challenges and opportunities each sector presents for business management	**BT 3**
CO 3	**apply** effective marketing and operational strategies to real-world food business scenarios to enhance performance and customer satisfaction.	**BT 3**
CO 4	**analyse** financial statements and performance metrics to assess the viability and profitability of food business ventures	**BT 4**

Course Outline:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Introduction to Food Business Management**: Importance of the food industry, key sectors in the food business: production, processing, distribution, and retail, business models in the food industry, understanding the food supply chain, legal and regulatory framework (FSSAI, CODEX, etc) for food businesses, ethical considerations in food management,	**15**
II.	**Marketing and Consumer Behaviour in the Food Industry:** Fundamentals of food marketing, understanding consumer behaviour and preferences, market research techniques in the food sector, segmentation, targeting, and positioning strategies, branding and product development in food businesses, pricing strategies for food products, promotional strategies: advertising, public relations, and sales promotions, digital marketing in the food industry,	**15**

	case studies of successful food marketing campaigns.	

III.	**Operations and Supply Chain Management in Food Businesses:** Overview of operations management in food production and service, quality management systems in the food industry, supply chain management concepts and practices, inventory management and control techniques, logistics and distribution strategies, sustainability in food supply chains, role of technology in food operations, food safety and traceability in supply chains, case studies on efficient food operations.	**15**
IV	**Financial Management and Entrepreneurship in the Food Industry:** Introduction to financial management principles in food businesses, budgeting and forecasting for food enterprises, financial statement analysis and key performance indicators, sources of funding for food businesses: loans, venture capital, and grants, entrepreneurship in the food sector: starting and managing a food business, business plan development for food ventures, risk management in food businesses, case studies of successful food entrepreneurs, exploring trends and innovations in the food industry.	**15**
TOTAL		**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential Learning:

1. Group Discussion
2. Seminar/presentation on any of the relevant topics

Case Studies

Texts:

1. David D and Erickson S. 1987. Principles of Agri Business Management. Mc Graw HillBook Co., New Delhi.
2. Acharya S S and Agarwal N L. 1987. Agricultural Marketing in India. Oxford & ISH Publishing Co., New Delhi.

References:

1. Cundiff Higler. 1993.Marketing in the International Environment, Prentice Hall of India,New Delhi.
2. Batra G S & Kumar N. 1994. GAD implications of Denkel proposals – Azmol PublicationsPvt., New Delhi.
3. Phill Kottler .1994. Marketing Management - Prentice Hall of India, New Delhi

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Level: Semester VIII

Course: Major **Scheme of Evaluation: (T)**

Title of the Paper: Nutraceuticals and Functional Foods
Subject Code: FTC152M801
L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To understand the Fundamental knowledge of nutraceuticals and functional foods and knowledge about health promoting effect of nutraceuticals and various metabolic disorders and their prevention and treatment by nutraceuticals and processing and storage of functional foods.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** knowledge of nutraceuticals and functional foods	**BT 1**
CO 2	**explain** the health promoting effect of nutraceuticals.	**BT 2**
CO 3	**explain** role of functional beverages and herbs and their market aspects	**BT 3**
CO 4	**apply** the role of functional beverages and herbs and their market aspects	**BT 4**

Course Outline:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Technological Aspects of Nutraceuticals and functional Foods:** Defining nutraceuticals and functional foods, nature, type and scope. Nutraceuticals and functional foods applications and their health benefits, classification based on chemical and biochemical nature with suitable and relevant descriptions.	**15**
II.	**Nutraceuticals for Specific Diseases:** Food recommended and restricted in metabolic disorders and disturbances, gastrointestinal disorders; fever and infection; liver, gall, bladder and pancreatic disturbances; blood circulatory and cardiac diseases; urinary and musculoskeletal diseases; allergies.	**15**
III	**Functional Role of Food Components:** Antioxidants, phytochemicals, isoflavones, lycopene, their role in Nutraceutical and functional foods, dietary fibers and complex carbohydrates as functional food ingredients. Proteins as a functional food ingredient, probiotic foods and their functional role.	**15**
IV	**Role of Specific Food Products as a Functional Food** Herbs as functional foods, health promoting activity of common herbs. Cereal products as functional foods- Oats, Wheat bran, rice bran etc. Functional vegetable products, oil seeds and sea foods. Coffee, tea and other beverages as functional foods/ drinks and their protective effects	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics

Texts:

1. Handbook of Nutraceuticals. Pathak YV, CRC Press, 2009
2. Innovations in Healthy and Functional Foods Ghosh D et al., CRC Press,2012

References:

1. Handbook of Nutraceutical and Functional Foods Wildman REC
2. Anti-angiogenic Functional and Medicinal Foods Losso JN

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Level: Semester VIII

Course: Major **Scheme of Evaluation:**

Title of the Paper: Concepts of Food Engineering

Subject Code: FTC152M702

L-T-P-C: 3-1-0-4 **Total credits: 4**

Course Objectives

To acquaint students with the fundamentals of food engineering and its process and to understand the basics of designing food plants and systems

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the basic laws of engineering.	**BT 1**
CO 2	**Explain** heat, mass and momentum transfer	**BT 2**
CO 3	**Apply** transport phenomenon in food processing operations	**BT 3**
CO 4	**Solve** problems related to transport phenomenon	**BT 4**

Course Outline

Modules	Topics (if applicable) & Course Contents	Periods
I	**Physical properties of Foods**: Methods of Estimation of Shape-Roundness, sphericity, roundness ratio, size, volume- platform scale method, density, specific gravity-apparatus, porosity and surface area	**15**
II	**Thermal Properties of Foods**: Definitions-Specific heat, enthalpy, conductivity and diffusivity, surface heat transfer coefficient. Measurement of thermal properties like specific heat, thermal conductivity and thermal diffusivity	**15**
III	**Aerodynamic properties and Frictional properties of Foods**: Aerodynamic Property- definition- drag coefficient, terminal velocity- application in handling and separation of food materials. Frictional property- coefficient of friction, angle of repose, angle of internal friction, application in food handling and storage	**15**
IV	**Rheology and texture of Foods**: Rheology-Rheological classification- viscoelasticity-viscometers, Hookean Body, St Venant body and Newtonian Body. Texture of foods-Methods of textural evaluation, Subjective and objective Method-texture profile method	**15**
	Total	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning

- Group discussion
- Seminar/presentation on any of the relevant topics
- Case studies

Textbooks

- Rao, M. A., Rizvi, S. S. H. and Datta A.K. latest edition. Engineering Properties of Foods: CRC Press.

- Heldman, D. R. latest edition. Food Process Engineering: AVI Publications.
- Toledo, R. T. latest edition. Fundamentals of Food Process Engineering (2 ed.): CBS Publications, New Delhi.

Reference Books

- Warren McCabe, Julian Smith, Peter Harriott. Unit Operations of Chemical Engineering. 7th Edition. McGraw Hill.
- R. Paul Singh, Dennis R. Heldman, and Ferruh Erdogdu. Introduction to Food Engineering.Latest Edition. Elsevier Science.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Level VIII

Course: Minor **Scheme of Evaluation: (T)**

Title of the Paper: Food Quality and Sensory Evaluation

Subject Code: FTC152N801

L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To understand knowledge of quality control and total quality management system in food industry and knowledge about food hygiene and importance of safe food and Food safety management system.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the concepts of food quality and role of total quality management system in food industry	**BT 1**
CO 2	**explain** the quality of food products using various techniques	**BT 2**
CO 3	**explain** the national & international food laws and regulations for quality of foods	**BT 3**
CO 4	**apply** to different techniques to perform sensory evaluation study of foods	**BT 4**

Course Outline:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Quality and Assurance:** Definition, scope, importance and difference, Total quality control and (TQC) Total quality management (TQM), Statistical quality control. Definition, importance, scope and difference between food quality and food safety.	**15**
II.	**Raw materials & Finished product quality**: Quality parameters and evaluation procedures: appearance, color, texture, viscosity, consistency, flavour etc. Food standards and laws: International – Concept of Codex Alimentarius, HACCP, GMP, GHP, USFDA, ISO 9000, ISO 22000, ISO 14000. Export Quality Control and Inspection act (1963), Environment Protection Act (1986), WTO & GATT, etc. GMP, GHP. National – Compulsory and voluntary trade and Company standards. Consumer Protection Act (1986), BIS/IS, Food Safety and standards – 2006, FPO, MPO, MMPO, Agmark. GMP, GHP.	**15**
III	**Quality Certification & Accrediation**: Introduction and procedure Prevention of food adulteration Act: Food Adulteration: definition, common adulterants in different foods, contamination, method of detection, Food additives and legislation; PFA specification for food products, Nutritional labeling Risk and Hazard associated with Food : Food hazards, sources of hazard, classification, Food safety; prevention and control, Statistical quality control. HACCP, Quality costs.	**15**
IV	**Sensory Evaluation:** Selection of panel of judges, Prerequisite for sensory analysis, application of consumer tests; control of factors affecting of sensory verdict, Instrumental measurements of sensory attribute of foods sensory characteristics of foods, types of tests, Texture profile analysis. Correlation between instrumental and Sensory analysis of food quality attributes.	**15**

	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics

Texts:

1. Food Quality Assurance: Principles and Practices, I. Ali, CRC Press, 2nd Edition, 2004.
2. Quality Assurance in Food Industry: a Practical Approach, J. A. Vasconcellos, CRC Press, 1st Edition, 2003.
3. Statistical Quality Control for the Food Industry, M. R. Hubbard, Kluwer Academic/ Plenum Publishers, 3rd Edition, 2003.

References:

1. HACCP in Meat, Poultry and Fish Processing, A. M. Pearson and T.R. Dutson, Kluwer Academic Publishers, 2nd Edition, 1999.
2. Food Safety Contaminants and Toxins, J. P. F. D'Mello, Oxford University Press, 1st Edition, 2003.
3. Quality of Fresh and Processed Foods, F.Shasidi, A.M.Spanier, Chi-Tang Ho and T.Braggins, Kluwer Academics/ Plenum Publishing,3rd Edition, 2004.
4. Rapid and On-line Instrumentation for Food Quality Assurance, Jotbill, Woodhead Publishing,2nd Edition, 2003.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Level VIII

Course: Major **Scheme of Evaluation: (T)**

Title of the Paper: Food Plant Sanitation

Subject Code: FTC152M802

L-T-P-E : 3-1-0-4 **Total credits: 4**

Course Objectives

Understanding the importance and process of sanitation in food processing facilities and learning of the sanitary processes and practices for production of safe and quality food products

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**explain** the importance of sanitation in food processing facilities	**BT 2**
CO 2	**identify** different contaminants of potential hazards in the food processing plants and equipment	**BT 3**
CO 3	**apply** the knowledge of hygienic design principles to avoid contaminants	**BT 3**
CO 4	**analyse** the food processing plants and methods to employ appropriate sanitation protocol and practices	**BT 4**

Course Outline:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Introduction to Food Plant Sanitation:** Definitions and importance of sanitation, Regulatory standards, Key terminologies (Clean, sanitize, disinfect), Microbial Contamination in Food Processing, Types of contaminants, Common foodborne pathogens, Sources of contamination, Types of soil contaminants- Organic, inorganic soils, Biofilms, Understanding allergens and cross-contamination, Cleaning and Sanitation Methods, Detergents and cleaning agents, Sanitizers, Physical methods	**15**
II.	**Hygienic Layout and equipment design**: Hygienic Design Principles: Layout considerations for sanitation, separation of clean and unclean areas, and hygienic design of walls, floors, ceilings, and drains, Sanitation in Equipment Design: Materials of construction (stainless steel, plastics), clean-in-place (CIP) systems, and proper equipment maintenance, Water and Waste Management: Water quality standards, waste management practices, and environmental impacts.	**15**

III.	**Sanitation Protocols and Practices:** Standard Operating Procedures (SOPs), Development, implementation, and monitoring of SOPs for cleaning and sanitation, Employee Hygiene, Personal hygiene practices, training, and compliance, Cleaning and Sanitizing Techniques: Proper use of cleaning agents, allergen control, and verification of sanitation.	**15**
IV	**New methods of sanitation and safety standards:** new cleaning technologies (e.g., UV light, ozone), Simple automation in sanitation: CIP and basic robotics, Sustainability in sanitation: Reducing chemical use and water consumption, Hazard Analysis and Critical Control Points (HACCP), Food Safety Audits: Types of audits (internal, external), audit procedures, and corrective actions, Certification Standards: ISO 22000, FSSC 22000, and SQF certification requirements	**15**
TOTAL		**60**
Pedagogy: Lectures, Assignments, Seminars		

Experiential Learning:

1. Group Discussion
2. Seminar/presentation on any of the relevant topics

Case Studies

Texts:

1. Roday, S. 1998. Food Hygiene and Sanitation, Tata McGraw-Hill Education.
2. Principles of Food Sanitation, 4th ed., Norman G. Marriott, 1999

References:

4. Sanitation in Food Processing, John A. Troller, 1993. Academic Press
5. Food safety management: A practical guide for the food industry. Lelieveld, H. L., & Motarjemi, Y. 2013. Academic Press.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**

Level VIII

Course: Major **Scheme of Evaluation: (T)**

Title of the Paper: Food Adulteration and Additives

Subject Code: FTC152M803

L-T-P-C: 4-0-0-4 **Total credits: 4**

Course Objectives

To aware students about basic idea on various foods and about adulteration and know about adulteration of common foods and their adverse impact on health and develop the skills of detecting adulteration of common foods and extend their knowledge for remedial measures for common food adulterants and knowledge of food additives and their applications in food processing

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**define** on various foods and about adulteration	**BT 1**
CO 2	**explain** adulteration of common foods and their adverse impact on health	**BT 2**
CO 3	**explain** the certain detecting adulteration of common foods	**BT 3**
CO 4	**apply** for food additives and their functions and regulatory aspects of food additives	**BT 4**

Course Outline:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I.	**Common Foods and Adulteration:** subjected to adulteration-Adulteration-Definition-Types, poisonous substance, Foreign Matter, cheap substitutes, spoilers part. Adulteration through food Additives	**15**
II.	**Adulteration in common Foods and methods of detection:** Means of adulteration, methods of detection adulteration in the following foods; milk, oil, grains, sugar, spices and condiments, process foods, fruits and vegetable	**15**
III	**Present Law and Procedures of Adulteration:** Food Safety and standards Authority of India -Rules and Procedures of Local Authorities, International – Concept of C ISO 9000, ISO 22000, ISO 14000. Consumer Protection Act (1986), BIS/IS, Food Safety and standards – 2006, FPO, MPO, MMPO, Agmark. GMP, GHP	**15**
IV	**Food Additives:** Definitions, sources, uses and functions and regulatory aspects of food additives and sweetening agents, Food additives: Use in fruit and vegetable preservation, food additives and legislation; PFA specification for food products, Nutritional labeling	**15**
	TOTAL	**60**
	Pedagogy: Lectures, Assignments, Seminars	

Experiential Learning:

- Group Discussion
- Seminar/presentation on any of the relevant topics

Texts:

1. Rapid Detection of Food Adulterants and Contaminants; Theory and Practice by Shyam Narayan Jha; Academic Press, 2016.
2. A first course in Food Analysis – A.Y. Sathe, New Age International (P) Ltd., 1999.

References:

1. Food adulteration and food fraud by Jonathan Rees, London, Reaktion Books, 2020.
2. Fennema's Food Chemistry by Damodaran Parkin Fennema, CRC Press, 2017.

Credit Distribution		
Theory	**Practical**	**Experimental Learning**
60	**-**	**30**